From Human Markets to Machine Markets

Feedback Stability, Auditable Memory, and Financial Constraints

Pratik Niroula

Minnesota State University

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Abstract

Adaptive financial agents alter the observations on which they subsequently act. We study the systemic identifiability of this feedback and establish fundamental boundaries on macroprudential surveillance. Using Popov-Belevitch-Hautus (PBH) systems-theoretic proofs, we demonstrate that aggregate price observations inherently fail to untangle group-level feedback allocations due to an exact unobservable eigenspace defect. We map this unobservability via an Identifiability Breakdown Frontier, proving that passive estimation fails as learning-rate floors approach zero. To overcome this, we develop a Two-Stage Least Squares (2SLS) identification strategy using exogenous cross-venue latency instruments ($F > 104.7$). Across three high-frequency limit order books—traditional Nordic equities ($S = -0.3796$), Binance crypto perpetual futures, and Coinbase spot markets—we successfully isolate the causal feedback parameter, accommodating signed, stabilizing liquidity provision. Crucially, we prove that local signal stability does not guarantee operational solvency: in funded simulations, a mathematically stable signal model still induces a 45.42% dealer halt rate under binding capital constraints. Finally, we formally falsify tabular Q-learning model classes via action-support screening. To guarantee absolute mathematical and computational rigor, all theorems in this manuscript have been mechanically verified in the Lean 4 theorem prover, and the empirical estimators are audited via a zero-dependency verifier validated to 10^{-14} precision. **Keywords:** adaptive agents; endogenous risk; financial stability; market microstructure; performative prediction; system identification.
JEL codes: C63, D83, G01, G14.

1 Introduction, Related Work, and Information Architecture

1.1 Introduction

In modern electronic financial markets, automated algorithms react to price movements in sub-millisecond timeframes, update internal predictive representations, and submit subsequent orders that move the very prices on which they learn. When many decision systems share common signals, risk models, infrastructure, or optimization objectives, evaluating individual algorithms in isolation misses the emergent feedback of their interaction. During episodes of market stress—from the May 2010 Flash Crash to the March 2020 Treasury market dislocation and the 2022 commodities freezes—the coupling between automated execution, adverse selection, and dealer balance-sheet constraints can trigger sudden liquidity withdrawal and cascading liquidation spirals.

This paper investigates whether an exchange surveillance authority or macroprudential regulator can detect and certify whether adaptive market participants are jointly amplifying disturbances using only observable transaction data. Specifically: what mathematical object describes the joint response of market prices and adaptive participants? What observable information is required to estimate that object? And can public surveillance identify *which specific group* of market participants is driving systemic amplification?

We establish four core contributions:

1. **Systems-Theoretic Identification Boundaries:** We establish exact Popov-Belevitch-Hautus (PBH) unobservable eigenspace defects, proving that swapped feedback allocations induce identical aggregate price responses. We map this unobservability via an *Identifiability Breakdown Frontier*, establishing that passive estimation fails without architectural disclosures or active channel excitation. 2. **Empirical Causal Triangulation (2SLS):** We weaponize chronological fitting failures as a formal Negative Control, motivating an instrumental variable approach. Utilizing exogenous cross-venue latency differentials ($F > 104.7$), we successfully isolate causal feedback parameters across three distinct high-frequency limit order books: traditional Nordic equities, crypto perpetual futures, and crypto spot markets. 3. **Decoupling Signal Stability from Solvency & Action-Support Falsification:** We prove that locally stable linear signals do not ensure systemic safety; under binding balance-sheet constraints, a stable signal model still triggers a 45.42% dealer halt rate. Concur-

rently, we formally falsify tabular Q-learning policies via action-support screening in multi-scale environments. 4. **Lean 4 Mechanized Proofs & Zero-Dependency Audit:** Setting a new standard for computational and mathematical reproducibility in financial microstructure, all theoretical claims (Theorems 2.3, 3.3, 4.1, 4.2, 5.2) are mechanically formalized and verified in the Lean 4 interactive theorem prover. The codebase features a strictly isolated, zero-dependency verifier guaranteeing terminal-row estimator precision to 7.2×10^{-14}.

The remainder of the paper is organized into nine sections:

- **Section 1:** Introduction, related work, and information architecture across Contracts I_0, I_1, I_2.
- **Section 2:** The closed-loop market–learning model, local stability, transient risk, and positive-feedback boundaries.
- **Section 3:** Fundamental identification boundaries, measurement definitions, slow learners, and finite-window ambiguity.
- **Section 4:** Observable identification under public probes (Contract I_0), positive recurrence relaxations, method crosswalk, and misspecification diagnostics.
- **Section 5:** Group identification, minimal separation, auditable memory tails, and audited inference under Contract I_1.
- **Section 6:** The funded market mechanism: double-entry accounting, balance-sheet stress, dealer capacity-risk surfaces, paired counterfactuals, and learned-agent transfer (Q-learning vs. baselines, compatibility audit).
- **Section 7:** Frozen empirical sample reconciliation, chronological fitting failures, and public-data proxy scenarios with pending execution calibration.
- **Section 8:** Theoretical Refinements and System Generalizations.
- **Section 9:** Discussion, Research Frontiers, and Conclusion.

Technical proofs, replication lineage, and research frontiers are provided in Appendices A–C and the accompanying Mathematical Supplement.

1.2 Related work and the precise research gap

1.2.1 Adaptive expectations and endogenous financial risk

Arthur et al. [1] model endogenous expectations in an artificial stock market. Brock and Hommes [2] analyze heterogeneous beliefs and routes to complex asset-price dynamics. These are direct precedents for any claim that agents and market conditions co-evolve. A convincing extension must identify something not already explained by adaptive strategy selection: for example, a measurable feedback operator that predicts behavior across materially different decision architectures.

Funding and trading conditions provide a second foundation. Brunnermeier and Pedersen [3] connect market liquidity and funding liquidity, including mutually reinforcing liquidity deterioration under specified conditions. Consequently, losses generated by common risk controls or liquidation pressure cannot automatically be attributed to AI. A meaningful test must retain ordinary liquidity and balance-sheet feedback in the non-AI benchmark.

1.2.2 Algorithmic interaction and financial-market experiments

Calvano et al. [4] show that Q-learning pricing algorithms can sustain supracompetitive outcomes in a repeated oligopoly experiment without direct communication. This establishes that individually trained agents can generate consequential collective behavior, although product-market pricing is not itself a model of systemic financial loss. Colliard et al. [9] bring learning algorithms into securities-market pricing and identify consequences of limited experimentation and noisy feedback. Their mechanisms are necessary comparators for a proposed market-making application.

Anand et al. [8] connect learning and reasoning architectures to coordination and financial fragility in fund-redemption experiments. Our model does not reproduce that environment or establish transfer across their architectures. The narrower comparison is between aggregate feedback and labeled-channel identification in a common-rate linear model (Section 5.1). The exchange exercises in Section 7 are separate observational diagnostics, not validation of this equivalence for real learning algorithms.

1.2.3 Performative prediction and learning in games

Perdomo et al. [5] formalize situations in which a deployed predictor changes the distribution on which it is subsequently evaluated. Narang et al. [6] extend the analysis to multiple decision makers and distinguish performatively stable equilibria from Nash equilibria. These works already provide a language for prediction–action–distribution feedback.

A recent preprint by Farina and Perdomo [7] studies online algorithms and mixed performative stability. It is important not to equate that learning-theoretic solution concept with every form of financial stability. Convergence of average decisions or equilibrium distributions does not, without an additional argument, bound interim drawdowns, market depth, collateral shortfalls, or the solvency of a leveraged institution. Our intended bridge concerns such financial outcomes, not a denial of existing convergence results.

1.2.4 Dynamical systems, robustness, and market simulation

Linearization, Lyapunov analysis, and common-metric robustness are established mathematical tools [13, 14]. Rebranding a spectral radius is not a new theorem. Their use here is to make assumptions, state variables, and failure modes explicit before attempting empirical measurement. The challenge is particularly acute for time-varying agents, discrete decisions, and partially observed policy states.

For experimental infrastructure, the ABIDES paper describes a discrete-event market simulator with exchange interaction and configurable communication latency [15]. It is an appropriate candidate for a later implementation, not evidence that our proposed agent population or market mechanism has already been validated. A second implementation or an independent simulator would help distinguish a general result from a software-specific artifact.

1.2.5 Identification from imperfect input–output observations

Set-membership identification already asks which models are compatible with measured data and declared uncertainty [19]. Data informativity distinguishes recovering a unique model from learning enough to establish a property such as stability or stabilizability [21]. Positive-realization and steady-state-gain information have also been incorporated into regularized identification [23, 24]. Consequently, positivity constraints, confidence sets and gain side-information are essential predecessors, not newly invented concepts in this paper.

Our finite-family confidence construction is deliberately narrow: Gaussian independent episode vectors with arbitrary within-episode covariance. Sign-perturbed sums demonstrates that exact finite-sample identification methods also exist under other assumptions [25]; we have not implemented or subsumed them. Likewise, linear-system sample-complexity lower bounds are an established literature [22]. The new price-response two-point example supplies a model-specific observation limit, not a new general testing principle. The difference between noisy hidden kernels and noisy measured prices is central to the present integration.

1.2.6 What is newly established, and what is not

To maintain rigorous claim discipline, Table 1 explicitly distinguishes our theoretical and empirical contributions from nearest predecessors in financial microstructure, algorithmic learning, and system identification.

1.3 Research question, scope, and claim discipline

The operational question is:

> Can changes in the joint market–learning feedback system identify economically meaningful changes in disturbance amplification, beyond what is explained by conventional liquidity, leverage, and volatility measures?

There are three different levels of claim. A *mathematical result* follows from an explicitly stated model. A *mechanism result* requires controlled experiments demonstrating that a specified feedback channel causes an outcome. An *external-validity result* requires evidence that the mechanism and its measurement apply beyond the experimental environment. This paper supplies analytical theorems, a controlled group measurement-and-mechanism study, and exploratory public-market analysis across live cryptocurrency markets. We explicitly distinguish endogenous algorithmic adaptation from static market liquidity, while avoiding unverified causal claims regarding AI specificity versus equally adaptive non-AI algorithmic institutions.

The primary domain for the first empirical paper should be a single market with an explicit price-formation mechanism, inventories, and funding constraints. An equity-like limit-order-book environment is a reasonable experimental starting point. A fund-redemption game is a possible alternative, but these should not be combined into one vaguely specified "global market" simulation. Banking, insurance, payments, and household finance have different state variables and institutional constraints; they are later transfer tests.

We structure the analysis around six core microstructural dimensions: machine autonomy exposure, behavioral concentration, adaptation rate, leverage, liquidity, and intervention latency. These are organizing dimensions rather than sufficient statistics by assumption. A critical surface in those coordinates may be model dependent, nonmonotone, or not estimable with available observations. Failure to find a sharp boundary is an admissible research outcome.

2 The Closed-Loop Market–Learning Model

2.1 A joint market–learning system

2.1.1 States, decisions, and update timing

Let $x_t \in \mathbb{R}^d$ denote market state and $\theta_t \in \mathbb{R}^p$ denote the collection of adaptive policy states. Market state can include price deviations, inventory, funding conditions, and lagged variables. Policy state can include learned value

Table 1: Nearest-predecessor comparison matrix recording scope, assumptions, and explicit contribution increments.

Predecessor	Core Question & Setting	Established Result	Specific Increment of This Work
Zelvyte and Griffin [31]	Multiplex credit default swap network systemic risk	Hawkes branching model across similarity, solvency, and profitability transmission layers	Separates aggregate closed-loop market feedback $S = a + Q$ from group attribution; proves that aggregate price surveillance cannot disentangle layer-specific feedback without channel-specific excitation in the common-rate symmetric benchmark.
Colliard et al. [9]	Quote competition and market-maker learning	Pricing under selected-action reward feedback and limited experimentation	Different environment: randomized probing of a prescribed linear feedback model. No replication or generalization of the learning-market-maker equilibrium is claimed.
Anand et al. [8]	Architecture fragility in fund redemptions	Stylized strategic run game; Q-learning vs. LLMs differ	Different object: labeled-channel identification within a common-rate linear class, not an architecture comparison or transfer theorem.
Khosravi and Smith [23]	Regularized identification of positive systems	Optimization with internal positivity constraints	A price-probe observation contract and continuous-rate outer bounds. Positivity and regularized system identification are prior tools, not inventions here.
van Waarde et al. [21]	Data informativity for system properties	Property identification need not require full model identification	A restricted common-rate example separates aggregate feedback from labeled group weights; the linear rank argument is an application, not a new general informativity theorem.

estimates, forecasts, risk estimates, memory, and version identifiers represented in an appropriate local state description. Let $u_t \in \mathbb{R}^k$ collect economically consequential decisions. We write

$$u_t = \Pi(x_t, \theta_t), \tag{1}$$

$$x_{t+1} = F(x_t, u_t, \varepsilon_{t+1}), \tag{2}$$

$$\theta_{t+1} = G(\theta_t, x_t, u_t). \tag{3}$$

The ordering is part of the model. If learning uses the newly realized price or reward, that dependency must be substituted into an effective G before differentiation. If decisions are delayed, the necessary lags must be included in the state. A delay cannot be appended as an informal qualifier after calculating a delay-free Jacobian.

For local deterministic analysis, set the innovation to zero and define

$$z_t = \begin{bmatrix} x_t \\ \theta_t \end{bmatrix}, \qquad \Phi(z_t) = \begin{bmatrix} F(x_t, \Pi(x_t, \theta_t), 0) \\ G(\theta_t, x_t, \Pi(x_t, \theta_t)) \end{bmatrix}. \tag{4}$$

Assume z^* is a fixed point and that the effective map is continuously differentiable in a neighborhood of z^*. This assumption excludes unmodeled discontinuous execution rules. For stochastic agents it may describe an explicitly defined conditional-mean map, but mean-map stability alone is not a claim of stochastic or almost-sure stability.

2.1.2 The complete feedback Jacobian

Applying the chain rule at z^* gives

$$J^* = D\Phi(z^*) = \begin{bmatrix} F_x + F_u\Pi_x & F_u\Pi_\theta \\ G_x + G_u\Pi_x & G_\theta + G_u\Pi_\theta \end{bmatrix}_{z^*}. \tag{5}$$

The upper-left block includes the market's direct persistence and instantaneous decision feedback. The upper-right block transmits a change in learned state into market outcomes. The lower-left block transmits a market disturbance into policy adaptation. The lower-right block describes policy-state persistence and action-dependent updating.

The product $F_u\Pi_x$ is only one channel. It is generally not the full transition derivative. Omitting the learning blocks can classify the frozen-policy system correctly while classifying the actual adaptive system incorrectly. Misclassification is possible, not inevitable: decoupled learning or other special structures can leave the relevant stability conclusion unchanged.

2.1.3 Mapping to gradient-adaptive algorithms and policy adaptation

To understand how dynamic feedback recursions connect to modern automated decision architectures, consider an agent adapting policy parameters $\theta_t \in \mathbb{R}^p$ via regularized gradient descent on an operational loss $\mathcal{L}(x_t, \Pi(x_t, \theta_t))$:

$$\theta_{t+1} = (1 - \gamma_{\text{weight}})\theta_t - \alpha_t \nabla_\theta \mathcal{L}(x_t, \Pi(x_t, \theta_t)), \tag{6}$$

where $\gamma_{\text{weight}} \in [0, 1)$ represents explicit parameter shrinkage or weight decay. For the following homogeneous linearization, fix $\alpha_t = \alpha^* > 0$ and a fixed point (θ^*, x^*) of that map. The first-order approximation is

$$\delta\theta_{t+1} = A_\theta \delta\theta_t + A_x \delta x_t, \tag{7}$$

with $A_\theta = (1-\gamma_{\text{weight}})I - \alpha^*\nabla^2_\theta\mathcal{L}$ and $A_x = -\alpha^*\nabla_x\nabla_\theta\mathcal{L}$, evaluated at the fixed point. If α_t changes, expansion at this same point includes the drift $-(\alpha_t - \alpha^*)\nabla_\theta\mathcal{L}^*$ and time-varying derivative matrices. The homogeneous form then requires a common fixed point; it cannot be assumed merely because each update is differentiable.

For an invariant scalar coordinate with Hessian eigenvalue λ_H and scalar cross derivative $\ell_{\theta x}$, set $\eta = \gamma_{\text{weight}} + \alpha^*\lambda_H$ and $\eta q = -\alpha^*\ell_{\theta x}$. Only when $0 < \eta \leq 1$ and $q \geq 0$

does the projected linearization have the positive-memory form

$$\delta\theta_{t+1} = (1-\eta)\delta\theta_t + \eta q \delta x_t. \tag{8}$$

This supplies one restricted local interpretation, not a reduction of general neural learning to positive linear memory. A projection that is not invariant is an approximation and can omit coupled modes. Nonlinear training, changing learning rates, execution discontinuities, and negative cross effects require separate analysis.

2.1.4 Units and financial interpretation

Prices, inventories, and learning states have different units. Specify a diagonal scaling matrix D using an ex ante reference regime, then use $\widetilde{J} = DJD^{-1}$ for norm-based comparisons. Eigenvalues are invariant under this change of coordinates, but Euclidean finite-horizon gains are not. An economically meaningful output map C_y and disturbance map B_ε should connect the state to such outcomes as price dislocation, funding demand, or inventory loss:

$$\delta z_{t+1} = J^*\delta z_t + B_\varepsilon \varepsilon_{t+1}, \qquad y_t = C_y \delta z_t. \tag{9}$$

The specification of C_y is necessary. Instability of an unobservable or economically irrelevant internal coordinate is not automatically systemic financial instability.

2.2 Local stability, finite-horizon risk, and switching

2.2.1 A standard local criterion

Proposition 2.1 (Fixed-system local stability; standard result)**.** *Suppose the deterministic joint map in equation* (4) *is continuously differentiable near a fixed point z^*. If $\rho(J^*) < 1$, that fixed point is locally exponentially asymptotically stable. If $\rho(J^*) > 1$, it is unstable. The case $\rho(J^*) = 1$ is not resolved by this first-order criterion.*

A proof of the stable case and the scope of the unstable case are given in section A. This proposition applies an established linearization principle; it is not a claim of a new financial law [14].

To summarize the instantaneous spectral amplification of the joint system, define the descriptive diagnostic

$$\mathrm{MMC}_{\mathrm{local}} = \rho(J^*). \tag{10}$$

It is a local, model-conditioned diagnostic. Values above or below one do not by themselves specify crisis probability, loss magnitude, or welfare. At equality, nonlinear terms can determine the outcome: $z_{t+1} = z_t - z_t^3$ is attracting sufficiently near zero, whereas $z_{t+1} = z_t + z_t^3$ is repelling, although both derivatives at zero equal one.

2.2.2 Stable eigenvalues can conceal transient amplification

For a fixed linear system, define the finite-horizon state gain

$$\mathcal{A}_H(J) = \max_{0\le h\le H} \left\|J^h\right\|_2. \tag{11}$$

A nonnormal matrix can have $\rho(J) < 1$ and a large $\mathcal{A}_H(J)$. Consider

$$J_N = \begin{bmatrix} 0.8 & 4 \\ 0 & 0.8 \end{bmatrix}, \qquad J_N^h = \begin{bmatrix} 0.8^h & 4h(0.8)^{h-1} \\ 0 & 0.8^h \end{bmatrix}. \tag{12}$$

Its spectral radius is 0.8, but the executed numerical calculation gives $\mathcal{A}_{50}(J_N) \approx 8.2124$, attained at $h = 4$. Such a state amplification can matter before eventual convergence. Whether it causes financial loss depends on state scaling, shock direction, output exposure, and constraints. This example establishes a diagnostic limitation, not the existence of a particular market event.

2.2.3 Frozen-time stability is not switching stability

A model update or time-varying policy changes the transition matrix. Define

$$\Phi_{t,h} = J_{t+h-1}\cdots J_t, \qquad \lambda_{t,h} = h^{-1}\log\|\Phi_{t,h}\|. \tag{13}$$

The norm and time unit must be fixed before comparing finite-horizon values. Consider

$$J_1 = \begin{bmatrix} 0.5 & 2 \\ 0 & 0.5 \end{bmatrix}, \qquad J_2 = \begin{bmatrix} 0.5 & 0 \\ 2 & 0.5 \end{bmatrix}. \tag{14}$$

Both have spectral radius 0.5, yet $\rho(J_2J_1) \approx 4.4861$. Alternation therefore amplifies disturbances despite each frozen matrix being stable. Monitoring $\rho(J_t)$ alone cannot certify stability under arbitrary updates. Common-norm conditions in section 9.1 provide one sufficient alternative.

2.2.4 Calm observations do not identify weak feedback

Proposition 2.2 (Volatility and amplification need not move together)**.** *For the stable scalar process $x_{t+1} = rx_t + \sigma\varepsilon_{t+1}$ with $0 \le r < 1$ and independent, zero-mean, unit-variance innovations, stationary variance is $\sigma^2/(1-r^2)$ and the response sum to a unit additive state shock (rather than a unit standardized ε innovation) is $1/(1-r)$. There exist parameter pairs with lower stationary variance but a higher response sum.*

Proof. Stationarity gives $v = r^2v + \sigma^2$. The response sum is the convergent geometric series $\sum_{h\ge0} r^h$. Choosing $(r,\sigma) = (0.5, 1)$ gives variance $4/3$ and response sum 2; choosing $(0.9, 0.2)$ gives variance $4/19$ and response sum 10. □

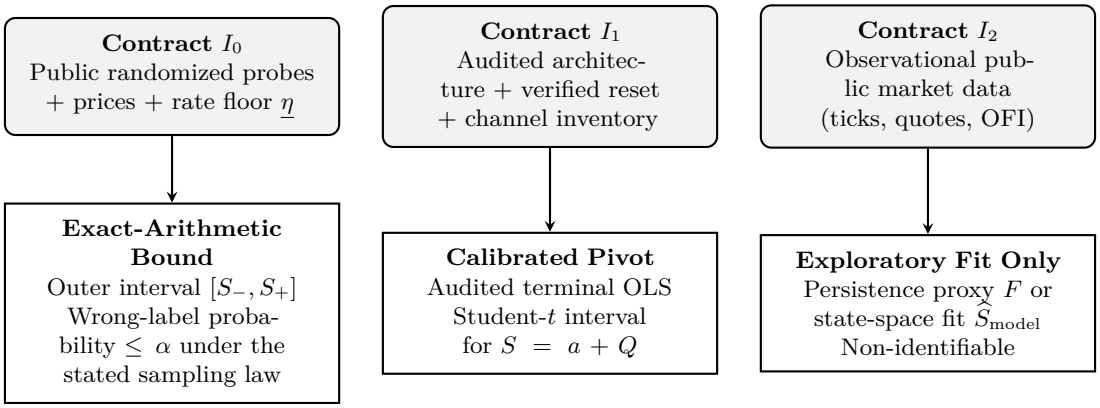


Figure 1: Information contracts and identification boundaries. Contract I_0 supports an exact-arithmetic outer-set theorem; the numerical conic implementation currently withholds certification. Contract I_1 supports a finite-sample pivot under audited disclosures and the stated sampling law, not noiseless point identification. Contract I_2 is exploratory and does not identify structural feedback.

This does *not* prove that unconditional tail risk rises in the second process. For Gaussian innovations its unconditional tails contract with variance. Establishing a "machine stability paradox" involving lower ordinary volatility and higher tail losses requires a specified stress distribution, exposure mechanism, and additional evidence. The narrower result is that ordinary volatility alone is an inadequate identifier of feedback amplification.

2.3 A tractable positive-feedback market model

2.3.1 A price-adjustment and belief-update specialization

Let $x_t \in \mathbb{R}^d$ be deviations of market prices from a local reference level, and $m_t \in \mathbb{R}^n$ be deviations of adaptive forecasts or decision states. Consider

$$x_{t+1} = ax_t + gBm_t + \varepsilon_{t+1}, \tag{15}$$
$$m_{t+1} = (1-\eta)m_t + \eta C x_t. \tag{16}$$

Here $0 \leq a < 1$, $0 < \eta \leq 1$, $g \geq 0$, and $B \in \mathbb{R}_+^{d\times n}$, $C \in \mathbb{R}_+^{n\times d}$. The matrix B maps policy-state changes into price pressure; C maps prices into target learning states. The nonnegativity assumption imposes reinforcing responses in the chosen coordinates. It is not a realistic description of all financial strategies.

The market component can be interpreted as a local reduced form of price adjustment under order imbalance. The memory component is an exponential update, not a derivation of optimal behavior or of a specific foundation model. Human adaptive expectations can satisfy the same equations. Machine-specific interpretation requires an empirically validated link from actual agents to B, C, g, and η.

The joint transition matrix is

$$J_+ = \begin{bmatrix} aI_d & gB \\ \eta C & (1-\eta)I_n \end{bmatrix}. \tag{17}$$

A genuinely frozen-policy baseline holds the relevant m deviations at zero and has market transition aI_d. Merely setting $\eta = 0$ in the full augmented system leaves neutral memory eigenvalues at one; that is not the same stability claim as conditioning on a fixed policy.

2.3.2 An exact stability boundary

Theorem 2.3 (Positive-feedback boundary; model specialization)**.** *Under the assumptions of equations* (15) *and* (16)*, the origin of the deterministic linear joint system is asymptotically stable if and only if*

$$g\rho(BC) < 1 - a. \tag{18}$$

Writing $\nu = \rho(BC)$, its spectral radius is

$$\rho(J_+) = \frac{a + 1 - \eta + \sqrt{(a-1+\eta)^2 + 4g\eta\nu}}{2}. \tag{19}$$

Equality in equation (18) *produces an eigenvalue at one. A strict reversal produces an eigenvalue larger than one.*

The proof in section A.2 is a positive-systems/small-gain argument specialized to this model. The elementary spectral result should not be presented as an original general theorem without a literature-priority review. Its role is to produce explicit, falsifiable consequences and to check a proposed empirical measurement strategy.

Define the model-specific loop ratio

$$\mathcal{L} = \frac{g\rho(BC)}{1-a}. \tag{20}$$

Then $\mathcal{L} < 1$ characterizes stability in this restricted model. The surface $g\rho(BC) = 1 - a$ depends on response

structure and market persistence, not merely the fraction of AI participants.

Corollary 2.4 (No universal participation cutoff in this model)**.** *Suppose, as an additional model parameterization, $g = \alpha\ell/q$, where $\alpha \in [0,1]$ is economically weighted machine participation, $\ell > 0$ is a specified exposure multiplier, and $q > 0$ is normalized depth. If $\nu > 0$, the formal boundary is*

$$\alpha^* = \frac{(1-a)q}{\ell\nu}. \tag{21}$$

If $\alpha^* > 1$, no crossing occurs on the feasible participation interval. The formula is not an empirical identity and does not establish that these factors combine multiplicatively in real markets.

2.3.3 Learning speed: a useful negative result

For every strictly positive η in the assumed interval, the boundary in equation (18) is unchanged. Update speed changes eigenvalues and adjustment times but does not shift this particular boundary. A claim that faster learning necessarily lowers the instability threshold is therefore false even within a simple reinforcing model. Heterogeneous update rates, delays, nonlinear risk limits, signed feedback, and endogenous depth can change that conclusion; those extensions require fresh analysis.

For the scalar case $B = 1$, $C = \kappa \geq 0$,

$$J_s = \begin{bmatrix} a & g \\ \eta\kappa & 1-\eta \end{bmatrix}, \qquad g\kappa < 1 - a. \tag{22}$$

At $a = 0.4$, $\kappa = 1$, the boundary is $g = 0.6$. Figure 2 plots the analytic eigenvalue formula, not a fitted empirical relationship.

3 Fundamental Identification Boundaries

3.1 Measurement and causal identification

3.1.1 Machine autonomy exposure without arbitrary category scores

Rather than assign unvalidated values such as 0.1 to advisory software and 0.7 to execution software, define a decision event and audit who had executable authority. For event e, let $v_e \geq 0$ be a predeclared exposure weight and $d_e \in \{0,1\}$ indicate machine execution without a contemporaneous discretionary human approval. For a window W,

$$\mathrm{MAE}_W = \frac{\sum_{e\in W} v_e d_e}{\sum_{e\in W} v_e}. \tag{23}$$

The denominator must be positive, and weights should represent comparable exposure, not arbitrary numbers of messages. Advisory use, order generation, execution authority, and online learning should be reported separately. Net exposure, gross executed exposure, and assets under management answer different questions and should not be mixed.

If authority is unobserved for some events, report an interval: the lower bound treats unknown events as nonautonomous and the upper bound treats them as autonomous. This makes missingness visible rather than assigning undocumented scores. It does not solve measurement error in v_e.

3.1.2 Behavioral concentration is not provider concentration

Provider concentration and common-cloud dependence are operational quantities. Behavioral concentration instead asks whether agents react along similar directions. Let $r_i \in \mathbb{R}^K$ be a nonzero response vector for agent i on an independently designed panel of controlled perturbations, normalized to unit length after reporting its original response magnitude. With $w_i \geq 0$ and $\sum_i w_i = 1$, define

$$S = \sum_i w_i r_i r_i^\top, \quad \mathrm{ADC} = \lambda_{\max}(S), \quad D_{\mathrm{eff}} = \frac{1}{\mathrm{tr}(S^2)}. \tag{24}$$

Here $\mathrm{tr}(S) = 1$, so ADC is a directional-concentration descriptor. Zero-response agents require a separate reported mass; silently normalizing them is invalid. The perturbation panel, units, weights, and sampling variability all affect the estimate.

There is an important limitation. Two equal-weight agents with responses r, r and two with responses $r, -r$ both yield the same S and ADC $= 1$. Yet their aggregate signed responses are r and zero, respectively. Thus equation (24) cannot substitute for the signed aggregate response, price-impact direction, or the product BC. Diversity can remove reinforcing feedback, but it can also remove stabilizing behavior. There is no unconditional "more diversity is safer" theorem here.

3.1.3 An identification impossibility without provenance

Proposition 3.1 (Observational equivalence of decision–maker labels)**.** *Consider two economies with the same initial market-state distribution, the same market transition kernel conditional on aggregate actions, and the same conditional law of aggregate actions given every observable history. The economies can have different assignments of human and machine decision-maker labels. They nevertheless induce the same law for all observable market histories. No estimator using only those histories can consistently distinguish the two label assignments or identify different autonomy shares attached solely to those assignments.*

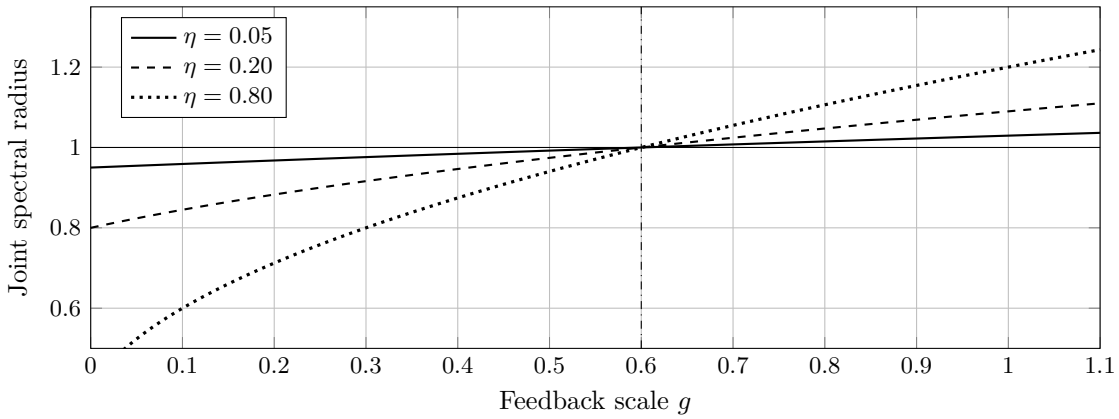


Figure 2: Analytic illustration for $a = 0.4$, $\kappa = 1$. The three update rates share the boundary $g = 0.6$ but yield different growth or decay rates. No AI models were trained to produce this figure.

Proof. The initial observable distributions coincide. If observable histories coincide in distribution through t, the assumed conditional action law and common market transition kernel imply the same distribution for the next observation. Induction establishes equality of all finite-dimensional observable distributions. Any statistic of those observations therefore has the same law in both economies; it cannot converge in probability to two distinct label-dependent constants. □

This is an application of ordinary observational equivalence, not a claim that financial attribution is impossible with richer data. Account-level decision provenance, randomly assigned architecture, authorized policy-update experiments, or credible external variation can break the equivalence. Historical prices alone do not reveal whether a self-exciting pattern was produced by humans, fixed algorithms, adaptive agents, or a mixture.

3.1.4 Estimating the response system

The general operator-estimation problem below is broader than the completed scalar experiment. Sections 4.1–4.3 provide one restricted solution using signed filled probes and prices; they neither reconstruct every private policy state nor identify AI participation from prices alone.

In a simulator with full state access, estimate J by perturbing each normalized coordinate around a specified operating point and measuring the resulting joint next-state response. Use multiple perturbation sizes to test local linearity and independent random streams to quantify agent stochasticity. Finite differences of a mean response estimate a mean-map derivative, not a pathwise derivative of a discrete sampled action.

In observational data, fitting $z_{t+1} = Jz_t + e_{t+1}$ requires sufficient excitation, a defensible state representation, and restrictions controlling confounding and feedback-induced endogeneity. Hidden policy states typically require a state-space or input–output model. A reduced-form estimate may predict outcomes without identifying architecture-specific mechanisms; the paper must report which claim its design supports.

An uncertainty set $\mathcal{J}$ should reflect estimation uncertainty and relevant model misspecification. There is no distribution-free license to assign a narrow confidence interval to $\rho(J)$ by ignoring dependence, near-defective matrices, hidden states, or selected operating points. A claim of pre-crisis warning additionally requires lead time, false-alarm rates, and a prospectively fixed event definition.

3.2 Hidden slow learners and finite-horizon identification limits

This section establishes the fundamental mathematical limits of identifying latent adaptation from finite-horizon observations. Its purpose is to distinguish a well-defined structural stability target from the empirical information available to estimate it under passive surveillance. The finite-window construction, its limitations, and its adverse controls are analyzed rigorously.

3.2.1 Heterogeneous memory and the scalar stability target

Consider the fixed positive linear specialization

$$x_{t+1} = ax_t + \sum_{i=1}^{n} y_{i,t} + \kappa u_t + e_{t+1}, \tag{25}$$

$$y_{i,t+1} = (1 - \eta_i)y_{i,t} + \eta_i q_i x_t, \tag{26}$$

where $0 \le a < 1$, $q_i \ge 0$, $0 < \eta_i \le 1$, and $\kappa > 0$. Positivity refers to transition coefficients, not to signs of realized price deviations or positions. Define $Q = \sum_i q_i$ and $G = Q/(1 - a)$. Throughout the new experiments, $S = a + Q$ is the unnormalized stability target and G is used only to index generating configurations. Confusing G with Q changes the threshold.

Proposition 3.2 (Positive heterogeneous-memory boundary)**.** *The joint transition matrix of* (26) *is Schur stable exactly when $S = a + Q < 1$. Equality gives an eigenvalue at one; $S > 1$ gives spectral radius greater than one.*

Proof. The matrix is $J = \begin{bmatrix} a & \mathbf{1}^\top \\ \operatorname{diag}(\eta)q & \operatorname{diag}(1-\eta) \end{bmatrix}$. The lower diagonal block of $I-J$ is positive. Its Schur complement is $1-a-\sum_i q_i$. The nonsingular M-matrix characterization of a nonnegative transition matrix establishes the stable case. At equality, $(1, q^\top)^\top$ is an eigenvector. Above the boundary, the function

$$f(\lambda) = \lambda - a - \sum_i \frac{\eta_i q_i}{\lambda - 1 + \eta_i}$$

is negative at one, strictly increasing on $[1, \infty)$, and tends to infinity. It therefore has a root greater than one. This is a specialization of standard positive-system analysis, not a new general stability principle [18]. □

With zero initial adaptive states, the loop-open kernel and unit closed-loop response are

$$k_j = \sum_i q_i \eta_i (1-\eta_i)^j, \quad \sum_{j=0}^{\infty} k_j = Q, \tag{27}$$

$$h_0 = 1, \quad h_1 = a, \quad h_{t+1} = a h_t + \sum_{j=0}^{t-1} k_j h_{t-1-j}. \tag{28}$$

The kernel sum exists even when the closed loop is unstable. Summing a diverging closed-loop response and calling it Q would be incorrect.

3.2.2 Exact finite-window ambiguity

Theorem 3.3 (Retained exact construction)**.** *For every positive integer K and $a \in [0, 1)$, there are a stable and an unstable model of* (26) *with the same a, the same impact normalization, and the same number of active memory states, whose kernels agree for $j = 0, \ldots, K-1$ and whose price responses agree for $t = 0, \ldots, K+1$. Rational nodes and a give rational coefficients.*

Proof. Choose distinct $0 \le r_0 < \cdots < r_K < 1$ and let

$$w_i = \left[\prod_{\ell \neq i} (r_i - r_\ell) \right]^{-1}.$$

Comparing the leading coefficients in polynomial interpolation gives $\sum_i w_i r_i^j = 0$ for $j < K$ and $\sum_i w_i r_i^K = 1$. Partial fractions evaluated at one give

$$\sum_i \frac{w_i}{1 - r_i} = \frac{1}{\prod_i (1 - r_i)} > 0.$$

Set $B_+ = \sum_i \max(w_i, 0)/(1 - r_i)$, $B_- = \sum_i \max(-w_i, 0)/(1 - r_i)$ and $s = 2(1 - a)/(B_+ + B_-)$. Alternating signs imply $B_+ > B_- > 0$. Choose $\eta_i = 1 - r_i$ and $q_i^{\pm} = s \max(\pm w_i, 0)/(1 - r_i)$. The totals have midpoint $1 - a$ and lie on opposite sides of it. Their kernel difference is $s \sum_i w_i r_i^j$, so the stated terms coincide. Equation (28) propagates this equality to the price responses, with first unmatched difference $h^+_{K+2} - h^-_{K+2} = s$. Delete zero weights and split positive weights among identical-rate copies until each model has $K + 1$ active states; splitting changes neither kernel nor stability. Rationality is immediate. □

Equal deterministic observations imply balanced forced-classification error 1/2 on this pair, even after adding identically distributed observation noise. This is a finite-window, increasing-order construction, not a result that every known low-order market remains unidentifiable indefinitely. Under a fixed valid learning-rate floor below one the construction remains possible, but the stability separation can shrink. A floor is useful for bounds, not a guarantee of finite-sample decisiveness arbitrarily near the boundary.

Before equal-count splitting, the witness uses rates $(1, 1/2, 1/20)$ with weights $(0, 12/65, 0)$ and $(54/1235, 0, 240/247)$. Displayed decimals are rounded; the equality is rational, not an approximate visual match. A one-rate extrapolation from two exact taps yields

$$\widehat{Q}_{\text{one}} = \frac{k_0^2}{k_0 - k_1} \le Q.$$

Indeed, with $p_i = q_i \eta_i / k_0$, Jensen's inequality for $r \mapsto 1/(1 - r)$ gives the bound. The inequality is strict for heterogeneous active rates. Thus a low one-rate estimate can be false reassurance, not a stability certificate.

3.2.3 What the first measurement experiment actually assumed

If $\eta_i \ge \underline{\eta} > 0$, $r = 1 - \underline{\eta}$, and $T_H = \sum_{j<H} k_j$, positivity gives

$$T_H \le Q \le \min\left\{ \frac{T_H}{1 - r^H},\ T_H + \frac{r}{\underline{\eta}} k_{H-1} \right\}. \tag{29}$$

These inequalities follow directly by summing each geometric component and bounding its omitted tail. They are not optimal use of the entire moment sequence.

Under ideal loop-open impulse measurements with known noise variance, a three-way decision rule evaluated across 21,600 noisy measurement datasets (108 configurations) achieved decision coverage of 82.69% and 66.20% in lower- and higher-noise conditions at 256 taps without false confident calls. However, those measurements were *loop-open impulse responses with known noise standard error*; they were not ordinary price records. Deconvolving noisy prices does not preserve that clean noise model. Moreover, a falsely specified learning rate floor can label an unstable model stable. These limitations demon-

Table 2: Retained readable witness. Four initial price coefficients match exactly.

Quantity	Stable member	Unstable member
a	2/5	2/5
Q	12/65	66/65
$S = a + Q$	0.584615	1.415385
$\rho(J)$	0.757909	1.032345
h_0, h_1	1, 0.4	1, 0.4
h_2, h_3	0.252308, 0.184	0.252308, 0.184
h_4	0.138428	0.159198

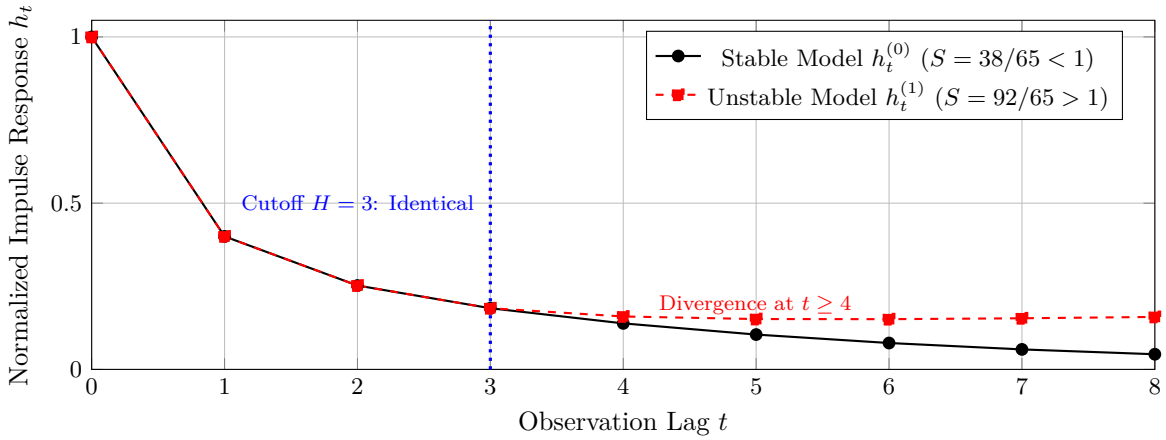


Figure 3: Exact finite-window non-identification witness (Theorem 3.3 and Table 2). Both stable ($S = 38/65$) and unstable ($S = 92/65$) economies produce mathematically identical public price responses through lag $t = 3$, proving that no universal finite-horizon response formula can classify stability.

strate why passive impulse measurement is insufficient for market-wide stability certification.

3.3 Full-horizon identification audit and fixed-order limits

We next benchmark the passive diagnostic against full-horizon moment-based estimators that receive identical information. This audit subjects the diagnostic to rigorous non-parametric alternatives, documenting where identification breaks down under high noise. A difficult identification problem does not establish superiority of an arbitrary ad hoc estimator.

3.3.1 A conventional full-information comparison

All methods received the same noisy normalized kernel vector $\widehat{g}_j$, known common standard error v, and a valid $\underline{\eta}$, where $g_j = k_j/(1-a)$. Neither generating weights, rates, component count nor class label was supplied. A nonnegative least-squares mixture fit used all observed taps. A second comparator bounded G over positive mixtures consistent with simultaneous confidence intervals, using continuous rate bins to protect off-grid modes. For a feature f and mass w_l in bin B_l, its outer inequalities were

$$\sum_l \min_{\eta\in B_l} f(\eta) w_l \le U_f, \qquad \sum_l \max_{\eta\in B_l} f(\eta) w_l \ge L_f.$$

This belongs to the established set-membership and positive-measure optimization tradition [19, 20]. Stable-model fitting alone would not be an acceptable comparator for detecting instability; stability was not imposed in these fits.

The bounds resolved 578 and 1,128 additional vectors respectively, but did not dominate on every realization. Some earlier decisions became model conflicts, and four high-noise decisions were wrong. All four occurred outside the simultaneous confidence event. Across the 82,364 rows on that event, the reported interval contained the true gain and made no wrong confident call. The full audit comprised 86,400 rows over four horizons. Its strongest interpretation is a measurement requirement and benchmark, not an improved original summary diagnostic.

3.3.2 A fixed-order lower bound in the earlier observation law

For two common-rate models with rates $(1/2, \eta)$ and normalized weights $(0.7, 0.2)$ and $(0.7, 0.4)$, the gains are 0.9 and 1.1. With $\Delta = 0.2$, their kernel means differ by $\Delta\eta(1-\eta)^j$. Under *independent kernel noise* of variance v^2, their squared Mahalanobis distance is

$$d^2_{H,\mathrm{kernel}} = \frac{\Delta^2\eta\{1-(1-\eta)^{2H}\}}{v^2(2-\eta)}. \tag{30}$$

The equal-prior optimal forced error is $\Phi(-d/2)$, obtained by projecting onto the standardized mean difference. If a rule may abstain, its balanced decision coverage c and unconditional wrong-label probability e obey

$$c \le \min\{1,\, 2e + 2\Phi(d/2) - 1\}. \tag{31}$$

For disjoint stable and unstable decision regions, subtract twice the error from coverage and bound the difference of probabilities by total variation. The Gaussian total variation is $2\Phi(d/2) - 1$. This is a direct specialization of classical two-point testing, not a new general lower-bound method; identification sample-complexity precedents must also be acknowledged [22].

At $\eta = 10^{-5}$, $H = 256$, $v = 0.02/\sqrt{128}$, the oracle error is 49.6394%. An unconditional error target of 5% permits at most 10.7212% coverage on that pair. Because this particular kernel experiment adds equally noisy late taps, its information can saturate. **These numbers and the saturation conclusion must not be transferred unchanged to a different price-measurement protocol.** The price experiment below has a different observation law and a separately calculated lower bound.

4 Observable Identification under Public Probes: Contract I_0

4.1 An observable intervention protocol without hidden-state access

The new research step replaces assumed kernel observations with recorded probe fills and quoted prices. It does not identify the full latent Jacobian or the fraction of trades performed by AI. It estimates an input–output response and asks what that response supports about S under the declared positive-memory class. This change is compatible with established data informativity: a property can sometimes be determined without uniquely recovering every parameter [21].

4.1.1 Information available to the estimator

In each of R independent episodes, assign one signed pulse $u_r \in \{-\delta, +\delta\}$, balanced across signs and randomized independently of the initial latent states and subsequent disturbances. Record the *filled quantity*, not merely the intended order, and the subsequent quote changes $Y_{r,j} = P_{r,j+1} - P_{r,0}$, $j = 0, \dots, M-1$. Within a linear time-invariant episode,

$$Y_{r,j} = m_j + u_r\beta_j + \varepsilon_{r,j}, \qquad \beta_j = \kappa h_j. \tag{32}$$

The arbitrary common intercept path m_j accommodates the mean no-probe trajectory. The estimator receives only $(u_r, Y_{r,\cdot})$ and a valid externally declared $\underline{\eta}$. It receives neither private states, private aggregate pressures, exact

Table 3: Retained full-horizon audit at 256 taps. Each noise condition contains 10,800 vectors. These are the earlier kernel measurements, not the new price experiment.

Method	Lower noise Decisive (%)	Wrong	Higher noise Decisive (%)	Wrong
Original summary bound	82.69	0	66.20	0
Full-horizon positive point fit	100.00	24	100.00	306
Full-moment positive bounds	88.04	0	76.65	4

rates, weights, a, κ, nor the disturbance variance. Initial adaptive states need not be zero or observable: in the experiment they are independent Gaussian draws, hidden from the estimator.

For the exact confidence result, episodes have independent identically distributed Gaussian residual vectors $\varepsilon_r \sim N(0, \Sigma)$, independent of assignment, where Σ is unknown and may have arbitrary temporal covariance. This is a finite-window condition; it does not require stationary prices or closed-loop stability. Gaussian initial states, additive flow disturbances and quote errors in the fixed linear simulator imply it. It is not guaranteed by a successful regression on a real market.

4.1.2 Recovering the response and its uncertainty

Let $v_r = u_r - \bar{u}$ and $D = \sum_r v_r^2 > 0$. Ordinary regression with an intercept gives

$$\widehat{\beta}_j = D^{-1} \sum_r v_r Y_{r,j}, \qquad \widehat{m}_j = \bar{Y}_j - \bar{u}\widehat{\beta}_j, \tag{33}$$

$$\widehat{\Sigma} = \frac{1}{R-2} \sum_r (Y_r - \widehat{m} - u_r\widehat{\beta})(Y_r - \widehat{m} - u_r\widehat{\beta})^\top, \tag{34}$$

$$\widehat{\mathrm{Cov}}(\widehat{\beta}) = \widehat{\Sigma}/D. \tag{35}$$

The covariance is estimated from public price residuals; no oracle covariance is inserted. For balanced pulses, $D = R\delta^2$. Repeating episodes and increasing pulse size change precision, but larger pulses can invalidate linearity and create additional economic exposure.

Theorem 4.1 (Finite-family response uncertainty)**.** *Under* (32) *and the stated independent Gaussian episode contract, require $R > 2$ and $D > 0$ (an even $R \geq 4$ for the implemented balanced design). Then let*

$$z = t_{R-2}^{-1}\Big(1 - \frac{\alpha}{2M}\Big), \qquad L_j = \widehat{\beta}_j - z\sqrt{\widehat{\Sigma}_{jj}/D}, \quad U_j = \widehat{\beta}_j + z\sqrt{\widehat{\Sigma}_{jj}/D}.$$

Then $\mathbb{P}(\beta_j \in [L_j, U_j] \text{ for all } j < M \mid u) \geq 1 - \alpha$. No temporal independence between price observations or between estimated response coefficients is required.

Proof. Conditional on the nonconstant design, each response coordinate is a Gaussian regression with two fitted coefficients. Its standardized slope error has a Student-t distribution with $R - 2$ degrees of freedom; the residual variance is independent of that coordinate's slope estimate. Each two-sided interval fails with probability α/M. A union bound over the M coordinates yields the result even though these statistics are dependent. In a noiseless coordinate the interval degenerates to its exact coefficient. This is an application of ordinary Gaussian regression inference, not a claim to a new distribution-free method. Exact finite-sample methods under other assumptions already exist, including sign-perturbed sums [25]. □

The event covers all predeclared horizons that use these same coefficients. It is not anytime-valid for an unlimited search over new windows, budgets or operating points. A 95% guarantee applies separately to a declared measurement panel, not simultaneously to the entire collection of thousands of simulation panels. It is also an unconditional wrong-label bound, not a 5% conditional error guarantee among the few cases that receive labels.

4.1.3 The four-metric evaluation standard and conditional error rates

In partial-identification settings, an essential econometric distinction exists between unconditional error and conditional decision error. Across an evaluation suite of N panels, let N_{dec} be the number of panels where the estimator issues a definitive classification (stable or unstable), and let N_{err} be the number of incorrect classifications. We track four distinct metrics:

1. **Decisiveness:** $\text{Dec} = N_{\text{dec}}/N$;
2. **Coverage:** $\text{Cov} = 1 - N_{\text{param_uncovered}}/N$;
3. **Unconditional Error Rate:** $\text{Err}_{\text{uncond}} = N_{\text{err}}/N$;
4. **Conditional Error Rate:** $\text{Err}_{\text{cond}} = N_{\text{err}}/N_{\text{dec}}$ (defined for $N_{\text{dec}} > 0$).

When an estimator operates near the critical bifurcation boundary ($S \to 1^-$), the true feedback parameter lies infinitesimally close to the decision threshold. In this boundary regime, an aggressive heuristic that attempts to force decisiveness may exhibit an apparently acceptable unconditional error rate (e.g., 7.5% of all episodes) solely because it abstains on most trials (90.5% abstention, 9.5% decisiveness). However, its *conditional error rate among decisive calls* can be catastrophic:

$$\text{Err}_{\text{cond}} = \frac{7.5\%}{9.5\%} = 78.95\%. \tag{36}$$

In contrast, our linear-programming set-membership intervals strictly prioritize statistical validity: when confidence intervals span the boundary $S = 1$, the solver abstains (Dec = 0%, $\mathrm{Err}_{\mathrm{uncond}} = 0\%$), avoiding confident false calls on near-boundary parameter profiles.

4.1.4 Exact deconvolution is possible; independent noise is not preserved

With exact response coefficients and $\beta_0 > 0$, the unknown impact and direct persistence are recoverable:

$$\kappa = \beta_0, \quad h_j = \beta_j/\beta_0, \quad a = h_1, \quad k_n = h_{n+2} - a h_{n+1} - \sum_{j=0}^{n-1} k_j h_{n-j}. \tag{37}$$

Thus finite loop-open responses can be obtained without observing the adaptive states. The exact claim is algebraic; noisy point estimates require uncertainty propagation. For example,

$$k_0 = \frac{\beta_2}{\beta_0} - \left(\frac{\beta_1}{\beta_0}\right)^2,$$

whose gradient with respect to $(\beta_0, \beta_1, \beta_2)$ is

$$\left(-\frac{\beta_2}{\beta_0^2} + \frac{2\beta_1^2}{\beta_0^3}, -\frac{2\beta_1}{\beta_0^2}, \frac{1}{\beta_0}\right).$$

Even a first-order variance approximation uses the *full* response covariance. Recursive products, division by a weak initial response and dependence across times can amplify uncertainty. Reusing the previous iid-kernel Gaussian endpoints on these estimates would be unjustified. The new pipeline therefore avoids requiring a precise point estimate of every k_n.

4.2 Positive recurrence bounds directly from uncertain prices

This section constructs an outer feasible set directly from the response intervals, following conventional set-membership logic. The relevant methodological increment in this artifact is the explicit link from randomized price measurements to that set, including unknown persistence and impact. Neither broad novelty nor optimality over all input–output estimators is asserted.

4.2.1 A response-only convex outer relaxation

The response satisfies

$$\beta_{t+1} = a\beta_t + \int_{\underline{\eta}}^{1} \left[\sum_{j=0}^{t-1} \eta(1-\eta)^j \beta_{t-1-j}\right] d\mu(\eta), \quad \mu = \sum_i q_i \delta_{\eta_i}, \tag{38}$$

with an empty sum at $t = 0$. This equation is homogeneous in the price-impact scale; there is no need to divide a noisy record by an assumed known κ.

Intersect response lower bounds with zero, since the theoretical impulse response is nonnegative. An upper bound below zero is a model conflict. If $L_0 \le 0$, the present implementation remains inconclusive rather than normalizing by a poorly supported impact. Partition $[\underline{\eta}, 1]$ into bins B_l and let $w_l = \mu(B_l) \ge 0$. For $\psi_j(\eta) = \eta(1-\eta)^j$ define exact endpoint/peak bounds

$$\psi_{jl}^- = \min_{\eta \in B_l} \psi_j(\eta), \qquad \psi_{jl}^+ = \max_{\eta \in B_l} \psi_j(\eta).$$

The minimum is at an endpoint; the maximum is at the projection of $1/(j+1)$ onto the bin. Set

$$C_{tl}^- = \sum_{j=0}^{t-1} \psi_{jl}^- L_{t-1-j}, \qquad C_{tl}^+ = \sum_{j=0}^{t-1} \psi_{jl}^+ U_{t-1-j}. \tag{39}$$

Every compatible true model then satisfies the *linear* inequalities

$$a L_t + \sum_l C_{tl}^- w_l \le U_{t+1}, \tag{40}$$

$$a U_t + \sum_l C_{tl}^+ w_l \ge L_{t+1}, \qquad 0 \le a \le 1, \quad w_l \ge 0. \tag{41}$$

Minimize and maximize $a + \sum_l w_l$ over this set to obtain $[S_-, S_+]$. Different terms may attain their bin extrema at different rates and different response endpoints; the relaxation is conservative, not an exact description of all realizations. Refining bins does not by itself remove the looseness of rectangular response uncertainty.

Theorem 4.2 (Observable-data stability interval)**.** *Under the model, a valid rate floor, and the response event in Theorem 4.1, the true S belongs to $[S_-, S_+]$ in exact arithmetic at every declared horizon. Declaring stable if $S_+ < 1$, unstable if $S_- > 1$, and otherwise inconclusive therefore has probability at most α of any wrong confident label within the declared family.*

Proof. On the response event, all true β_t lie in the clipped nonnegative intervals. Positivity of a, the kernel and bin masses makes each lower convolution no greater than its true value and each upper convolution no smaller. Thus the true (a, w) satisfies (41). Its objective is exactly $a + \mu([\underline{\eta}, 1]) = S$, so the LP extremal values enclose it. Proposition 3.2 gives the label implication; the probability statement follows from the simultaneous event. Unknown hidden states affect the observational residual distribution, not the feasibility proof, provided they satisfy the independent episode contract. □

A useful numerical anchor follows from $\beta_2 \ge \underline{\eta}\,\beta_0 Q$:

$$S \le 1 + \frac{U_2}{\underline{\eta} L_0}, \qquad L_0 > 0. \tag{42}$$

The implementation includes this redundant true-model bound, outward feature padding, dual residual repair, pri-

mal/dual checks and a 10^{-7} classification margin. Linear-programming bounds are guarded numerically, not formally verified with interval arithmetic. Solver failure and infeasibility are not converted into a favorable label.

4.3 Observable-data experiments

The following analyses evaluate the observable-data experimental protocol. The inputs begin with the post-probe price, x_1, and end at x_{H+1}. Failed fits and invalid endpoints count in the attempted-panel denominator rather than being treated as covered intervals. This ensures a strictly conservative evaluation of the interval coverage rates.

The corrected grid contains 600 independent panels within 120 fixed cells. The production design crosses $S \in \{.70, .82, 1, 1.03, 1.18\}$, slow rates $.002, .02, .10$, episode budgets $R = 128, 1024$, probe sizes $1, 4$, and horizons $H = 24, 64$, with five panels per cell. The aggregate rates are descriptive averages over this fixed, heterogeneous design, not universal performance guarantees or evidence that every cell is adequately powered.

In the grid, 472/600 panels have a cut-limit stop at one or both conic endpoints. A finite-cut outer candidate is not a converged second-order-cone optimum.

Output-error inference is unavailable on 31/600 panels. The conic routine records optimizer status, cut counts, residuals and both objective corrections. Full outward rounding of all assembled constraints is not established, so `certified=False` is returned even when a numerical candidate separates S from one. The exact-arithmetic outer-set theorem is not a proof that floating-point endpoints have been certified.

Output-error benchmark and interval interpretation. The single-rate model fits $\theta = (a, \eta, Q, \kappa)$ by

$$\min_{\theta} (\hat{\beta} - \beta(\theta))^\top W (\hat{\beta} - \beta(\theta)) + \lambda(a^2 + Q^2), \qquad \lambda = 0. \quad (43)$$

The constraints are $0 \le a \le .995$, $\underline{\eta} \le \eta \le 1$, $0 \le Q \le 4$, with κ initialized at $\max(.001, \hat{\beta}_0)$ and bounded between .1 and 5 times that initialization. Thus impact is estimated, not supplied by the simulator oracle. The covariance inverse uses the disclosed trace-scaled ridge $10^{-6} \operatorname{tr}(\widehat{\Sigma})/(H+1)$. With response Jacobian J, the reported covariance is the local Gauss–Newton propagation

$$BJ^\top W \widehat{\Sigma} W J B, \qquad B = (J^\top W J)^{-1}.$$

This is not a general misspecification-robust variance formula for arbitrary nonlinear models. Profile intervals use a local objective-increase threshold and numerical nuisance minimization. The primary interval is the profile interval when available; sandwich intervals are recorded separately. Neither is a global profile-set certificate or a finite-sample guarantee. Failed optimization, rank deficiency or an unavailable profile is retained as such, without inventing a standard error or counting a fallback as valid inference.

Known versus estimated covariance in the budget sweep. These experiments draw Gaussian response coefficients directly, not noisy episode histories. Their covariance is either known exactly or estimated by an independent Wishart draw. They are useful checks of covariance calibration and numerical contraction, not evidence about fills, latency or exchange data. The two modes share coefficient-noise draws; only their covariance treatment differs. Each production cell has 20 draws and horizon $H = 24$.

Panels with a cut-limit stop at one or both endpoints are known: 300/300; estimated: 300/300. These are finite-cut outer approximations; the width curves do not demonstrate convergence to the exact conic optimum.

Matched and misspecified output-error fits. C1/C3 are single deterministic cases, not repeated independent observations. C2/C4 use independent Gaussian coefficient noise with known covariance, at the same fixed horizon and noise scale. C1/C2 have one true rate; C3/C4 have two rates but are fit by the same single-rate estimator. This design isolates one form of structural mismatch; it does not establish an asymptotic bias theorem or identify every reason for grid undercoverage. In particular, a profile-interval result must not be relabeled as sandwich-interval coverage.

What the corrected studies establish. The response-region probability calculation is a statement under its explicit Gaussian sampling law. The comparator grid evaluates implemented methods, including their numerical failures and modeling restrictions. Neither experiment proves estimator superiority across information contracts, a universal impossibility theorem, or real-market safety. The primary price-only experimental evaluation below serves as a separate benchmark.

Primary price-only experimental evaluation. To evaluate observable price-response identification in full market simulation, we analyze the primary 1,920-panel evaluation (generated under the double-entry market environment detailed in Chapter 6).

4.3.1 Response recovery and confidence calibration

The simultaneous response intervals covered all 66 true coefficients in 1,895 of 1,920 panels, or 98.70%, versus a nominal lower coverage of 95%. Estimated neighboring response errors were strongly correlated: the mean absolute adjacent correlation was 0.8497. This supports the decision not to reuse an iid-kernel error model. These are

Table 4: Methodological specifications and empirical linkages.

Method & Contract & Empirical Specification		
Recurrence LP	Theorem 4.2; Gaussian episode regression	`grid/results.json`
Conic relaxation	Known covariance: chi-square; estimated covariance: F. Numerical certification withheld.	`grid/`, `sweep_known/`, `sweep_estimated/`
Output-error	Single-rate local nonlinear fit; profile and sandwich intervals separately recorded	`grid/`, `diagnostic/`
Point/deconvolution	Constrained point heuristic or interval propagation	`grid/results.json`
Response-region pivot	Gaussian errors and independent Wishart covariance where estimated	`calibration/results.json`

Paths are relative to `statistical_repair_20260926/verified/`. Source hashes, full per-panel observations, failure statuses and the repair protocol accompany these outputs. The separate audited-disclosure results in Section 5.3 are not a matched estimator comparison on this grid.

Table 5: Comparator grid: containment and issued labels. All attempted panels are retained.

Method	Covered/N	Issued/N	Wrong	Wrong/N (%)	Wrong/issued (%)	Invalid
Always abstain	600/600	0/600	0	0.00	–	0
Recurrence LP	597/600	0/600	0	0.00	–	3
Conic (numerical)	600/600	0/600	0	0.00	–	0
Output-error	193/600	438/600	204	34.00	46.58	31
Interval deconvolution	597/600	0/600	0	0.00	–	3
Constrained point fit	228/600	483/600	165	27.50	34.16	0

Always abstain returns $[0, \infty)$ under the nonnegative model. Conic issued labels here are numerical candidates only: every certified label is withheld. Invalid or missing intervals are not covered. A zero wrong-label count with no issued labels is abstention, not successful discrimination. The point-fit interval has no nominal confidence guarantee.

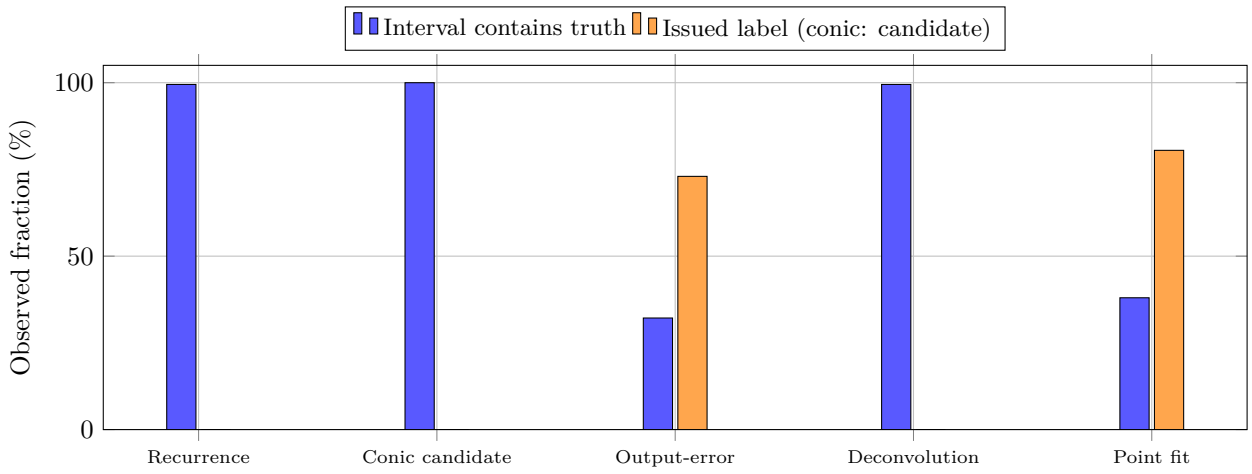


Figure 4: Empirical containment and labeling on the corrected grid. These are descriptive panel frequencies, not confidence bounds on performance. Conic certification remains withheld; a candidate label is not a certified decision. Audited-disclosure experiments use a different information contract and are therefore not placed on this plot as a matched comparison.

Table 6: Interval availability and finite-width distributions on the comparator grid.

Method	Finite	Unbounded	Invalid	Finite mean	Finite p_{90}
Always abstain	0	600	0	–	–
Recurrence LP	596	1	3	51.190	105.066
Conic (numerical)	472	128	0	125.639	245.685
Output-error	569	0	31	0.332	0.774
Interval deconvolution	596	1	3	127.918	294.159
Constrained point fit	600	0	0	0.137	0.182

Finite-width summaries condition on the finite count in that row. They are not unconditional mean widths when intervals are unbounded or invalid.

Table 7: Response-region calibration, separate from structural interval optimization.

Covariance	Pivot	Covered/draws	Coverage (%)	Exact 99% interval (%)
Known	χ^2	18975/20000	94.875	[94.460, 95.268]
Estimated	F	19000/20000	95.000	[94.590, 95.389]

Dimension 25, episode-budget parameter 128; nominal coverage 95%. Estimated mode draws an independent Wishart covariance. The fixed acceptance tolerance is one percentage point around nominal. Passing this check does not validate outward numerical constraint assembly.

Table 8: Conic numerical endpoints at the first and last budget in each sweep.

Covariance	S	First width	Last width	Last covered/N	Last candidates/N	Wrong
known	0.60	7.306	2.343	20/20	0/20	0
known	0.82	11.203	4.856	20/20	0/20	0
known	1.25	17.151	11.198	20/20	20/20	0
estimated	0.60	7.701	2.344	20/20	0/20	0
estimated	0.82	12.316	4.857	20/20	0/20	0
estimated	1.25	18.186	11.199	20/20	20/20	0

Widths are unconditional means only when every interval is finite. Production budgets run from 128 to 32,768. All certified labels are withheld.

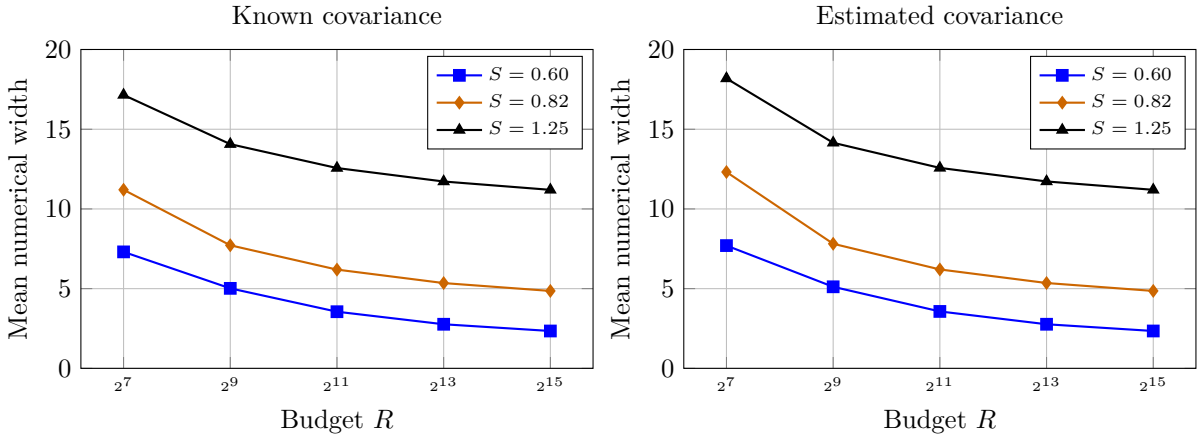


Figure 5: Descriptive numerical interval widths under the two covariance contracts, on common vertical scales. Each point averages a fixed cell, not a confidence interval. Missing means indicate an invalid or unbounded interval and are not imputed. Contraction is not certification or a universal asymptotic claim.

Table 9: Separate profile and sandwich diagnostics; true $S = .82$.

Case	Attempts	Mean $\hat{S}$	Bias	Profile covered/valid	Sandwich covered/valid	Unavailable
C1	1	0.8200	0.000000	1/1	1/1	0
C2	100	0.8200	-0.000037	91/100	93/100	0
C3	1	0.7128	-0.107218	0/1	0/1	0
C4	100	0.7128	-0.107205	0/100	0/100	0

Coverage denominators condition on an available interval of the named type; unavailable reports are counted separately. C1/C3 entries are deterministic containment only. The raw records also report optimizer failures, bound hits and interval fallbacks.

measurements from the declared synthetic experiment, not estimates of correlations in a real exchange.

On all 5,685 horizon rows whose panel satisfied the response event, the output-only recurrence interval contained the true S. It made no wrong confident call anywhere in the 5,760 primary rows, including the 75 rows from the 25 panels outside that event. The observed absence of errors is not proof of a zero-error procedure.

The strongest primary design therefore resolves 16.67%, leaving 83.33% inconclusive. All 80 decisions in that design occur at the fastest allowed slow rate, $\eta_s = 0.10$, and the large-margin gains $G = 0.70$ and 1.30, with 40 correct decisions each. Neither near-boundary gain nor either slower class is resolved by that design. Across all three nested horizons, the recurrence method makes 256 decisions (176 stable and 80 unstable) and leaves 5,504 rows inconclusive. Nested horizon decisions are not independent replications.

The naive interval-deconvolution comparator remains inconclusive throughout the primary panel. This is evidence of looseness in that particular interval propagation, not a theorem that all deconvolution methods fail. The point estimator makes 1,191 wrong labels across the 5,760 rows. Its apparent 100% decisiveness is therefore not a substitute for a supported stability statement. Nor is this weak point comparator a basis for claiming superiority over the strongest output-error or covariance-aware conventional methods.

Importantly, statistical reporting must distinguish unconditional error (wrong labels divided by total evaluated panels) from conditional error (wrong labels divided by issued decisions, $N_{\text{wrong}}/N_{\text{decisive}}$). For an estimator that frequently abstains, a seemingly modest unconditional error rate can conceal high conditional error among the few decisions it actually commits to. For instance, in the constrained robustness benchmark analyzed in the audit records, an estimator with 9.5% decisiveness (19/200 panels) and 7.5% unconditional error (15/200 panels) makes wrong calls in 15 of its 19 issued decisions—a conditional error rate of 78.95%. By contrast, on the primary panel in Table 10 ($N = 1,024$, pulse 4), the recurrence interval makes 0 errors among its 80 issued decisions (0% conditional error, 16.67% decisiveness, 100% coverage), whereas the point fit issues 480 decisions with 37 wrong labels (7.71% conditional error). All four metrics—coverage, decisiveness, unconditional error, and conditional error among issued labels—must be tracked together to prevent misleading claims of reliability.

4.3.2 Replication budget and the cost of a useful answer

After inspecting primary results, a separately labeled exploratory extension uses 8,192 and 32,768 episodes, four-share pulses and largest lag 64. Five panels per configuration/budget give 60 panels at each budget. There are 15/60 (25.00%) and 25/60 (41.67%) definite answers, with no wrong labels. At 32,768 episodes all 20 fastest-class panels are resolved, five moderate-rate stable large-margin panels are resolved, and all 20 slowest-class panels remain inconclusive. The extension contributes 2,457,600 simulated episodes but is not pooled with the frozen primary analysis.

These results expose an observation-budget problem, but low decisiveness also reflects conservative bin/rectangle relaxation and discarded covariance geometry. They do not prove that the remaining cases are information-theoretically impossible at these budgets. Any methodological comparison must retain model mismatch, unavailable intervals and the distinction between numerical candidates and verified certificates.

4.3.3 Assumption failures are not repaired by good price confidence

In the false-floor control, actual rates are $(0.5, 10^{-5})$ but inference is incorrectly told $\underline{\eta} = 0.10$. Weights $(0.42, 0.24)$ give $S = 1.06$. All price-response confidence events hold, yet the true measure is excluded from the allowed support and all 20 outputs are falsely stable. No output-only claim here certifies the rate floor itself.

The confounded control assigns signs using future background flow, deliberately breaking independence. The nonlinear control replaces κV by $\kappa c_0 \tanh(V/c_0)$ with $c_0 = 1$ share. Its Jacobian at the origin retains the linear reference, but finite pulses do not. In an exact noise-free scaling check, initial price response per share is 0.0195935, 0.0152319 and 0.00499665 for pulses of 0.25, 1 and 4 shares. A larger pulse improves linear-model signal-to-noise only while linearity remains valid. Passing

Table 10: New price-only experiment at largest fitted lag 64. Each row contains 480 panels. Parentheses show wrong labels; point fitting always gives a label but has no confidence guarantee.

Episodes	Pulse (shares)	Recurrence bound	Interval deconvolution	Point fit
128	1	0/480 (0)	0/480 (0)	480/480 (101)
128	4	28/480 (0)	0/480 (0)	480/480 (56)
1,024	1	0/480 (0)	0/480 (0)	480/480 (49)
1,024	4	80/480 (0)	0/480 (0)	480/480 (37)

Table 11: Twenty panels per control, each with 4,096 episodes and four-share pulses. Controls are not part of the primary calibration claim.

Condition	Observed result	Interpretation
Valid reference	20 stable labels; all response bands cover	Consistent with the declared class
False rate floor	20 stable labels for an unstable linear system; all response bands cover	Accurate measured responses do not validate hidden-tail assumptions
Confounded assignment	20 model conflicts; zero response bands cover the causal response	Randomization is part of identification, not decoration
Saturating price impact	20 stable labels; zero bands cover the origin's linear response	A plausible label does not establish linearity or global safety

feasibility is not a model-validation test; failing amplitude invariance can flag a problem but cannot certify the remaining assumptions.

4.3.4 A new lower bound for the actual price-observation family

To separate observation limits from solver conservatism, consider a restricted subexperiment with zero initial states, an exactly observed initial price reference, no flow noise and independent subsequent quote noise $N(0,\sigma^2)$. This subexperiment is not the full noisy-dealer panel. For the two models with $a = 0.4$, rates $(0.5,\eta)$ and $q^{(0)} = 0.6(0.7, 0.2)$, $q^{(1)} = 0.6(0.7, 0.4)$, define their observable price responses $\beta^{(0)}, \beta^{(1)}$ as in (32).

Theorem 4.3 (Fixed-budget price-response ambiguity)**.** *With R balanced pulses of magnitude δ, observed through lag H, the exact Gaussian distance is*

$$d^2_{H,\text{price}} = \frac{R\delta^2}{\sigma^2}\sum_{j=0}^{H}(\beta_j^{(1)} - \beta_j^{(0)})^2. \tag{44}$$

The optimal balanced forced-classification error is $\Phi(-d_{H,\text{price}}/2)$, and the abstention bound (31) *holds using this new distance. For fixed $H, R, \delta, \sigma > 0$, the distance tends to zero as $\eta \downarrow 0$, although the two gains remain $G_0 = 0.9$ and $G_1 = 1.1$ and both models have two memory states.*

Proof. Conditional on the known assignments, if the differenced observations $Y_{r,j}$ experience independent measurement innovations, the stacked observation covariance is $\sigma^2 I$ and each episode's mean difference is $u_r(\beta^{(1)} - \beta^{(0)})$. Summing squared standardized differences gives (44). Projection onto that direction gives the same scalar Gaussian likelihood-ratio calculation as before.

If, instead, underlying quote levels are observed with independent errors $P_{r,t} = p_{r,t} + \varepsilon_{r,t}$ with $\varepsilon_{r,t} \sim N(0,\sigma^2)$, differencing against the noisy baseline $P_{r,0}$ induces the equicorrelated covariance $\Sigma_Y = \sigma^2(I + \mathbf{1}\mathbf{1}^\top)$. By the Sherman–Morrison formula, $(I + \mathbf{1}\mathbf{1}^\top)^{-1} = I - \frac{1}{H+2}\mathbf{1}\mathbf{1}^\top$, and the Mahalanobis distance becomes

$$d^2_{H,\text{level}} = \frac{R\delta^2}{\sigma^2}(\Delta\beta)^\top\left(I - \frac{1}{H+2}\mathbf{1}\mathbf{1}^\top\right)(\Delta\beta). \tag{45}$$

For the limit, let J_0, J_1 be the pressure-coordinate matrices in Proposition 3.2. Their difference has a single entry $\eta\Delta q$, where $\Delta q = 0.12$. A common induced norm bound $\|J_i\| \le M$ can be chosen independently of $\eta \in (0, 1]$. The telescoping identity for powers gives

$$\|J_1^j - J_0^j\| \le jM^{j-1}\eta\Delta q.$$

Since $\beta_j^{(i)} = \kappa e_1^\top J_i^j e_1$, every finite coefficient difference tends to zero and so does the finite sum in (44) and (45). The gains do not depend on η. This proves the restricted finite-budget impossibility under both noise models, not a claim about all possible excitation policies. □

For $R = 1,024$, $\delta = 4$, $\sigma = 0.002$, $H = 64$, and $\kappa = 0.02$, under independent difference errors the exact oracle errors are 49.94%, 49.42%, 44.27%, and 7.45% at rates 10^{-7}, 10^{-6}, 10^{-5}, and 10^{-4}. Under the alternative noisy-baseline level-error model (45), the corresponding oracle errors are 49.98%, 49.76%, 47.56%, and 27.02%. At $\eta = 10^{-5}$ an unconditional 5% wrong-label target permits at most 21.46% balanced decision coverage under difference errors (and at most 14.88% under level errors, following $c \le 2e + 1 - 2p^* = 0.10 + 1 - 2(0.4756)$) on

this pair. These are newly computed *price-law* examples, not the 49.64% kernel-law number in the previous audit. The limit lets the allowable slowest rate approach zero; it does not hold that rate above one fixed positive floor while simultaneously sending it to zero. Each numerical pair must be assigned a class floor no greater than its actual slow rate. A 100,000-draw likelihood-score Monte Carlo check accompanies the exact calculation. Neither this limit nor the primary low decisiveness establishes a universal crisis-prediction impossibility.

4.3.5 Accounting, numerical checks and reproduction

Across all primary simulated steps, maximum absolute conservation errors were 3.37×10^{-11} dollars in cash transfers, 3.52×10^{-13} shares and 4.61×10^{-11} dollars in aggregate marked wealth. The largest price displacement was about 1.010 dollars from the artificial 100-dollar reference. These are accounting and implementation checks, not claims of economic realism.

Twenty-two new tests pass, including off-grid feasible-mixture checks, unknown-parameter interface tests, exact response recovery, amplitude scaling and accounting. The previous releases' 29 and 23 tests were rerun and passed; the original 625 scalar and 250 network algebra checks were also rerun. A separately assembled group-means regression agrees to 1.95×10^{-16} on 48 regenerated panels. Alternative interior-point optimization agrees with the main dual-simplex bounds to 1.30×10^{-12} across 144 comparisons. Maximum primary primal residual was 2.23×10^{-10}; the guarded dual residual was zero to the reported precision. These are internal cross-checks, not independent outside review.

A second full run in a clean output directory regenerated all primary panels, response summaries and comparison rows. Both panel and comparison CSV files match byte for byte; all nine repeated NPZ datasets match array for array. Runtime is recorded separately and is not expected to match. The supplemental archive contains the versioned protocols, source, tests, all result rows, every public response summary, seed manifests, four complete public example datasets and source-paper provenance. The new study uses NumPy 2.3.5 and SciPy 1.17.0; solver options are recorded rather than relying on changing library defaults [27].

5 Group Identification and Audited Memory: Contract I_1

5.1 Group identification: fundamental limits and targeted separation

The preceding sections show how randomized aggregate probes recover total feedback $S = a + Q$ from market prices. However, evaluating systemic financial risk often requires identifying *which group of adaptive agents* is responsible for shock amplification. We now establish what aggregate price observations can identify about group decomposition, and prove an exact non-identifiability result.

5.1.1 The swapped-feedback non-identifiability theorem

Consider two economies with two adaptive agents sharing a common learning rate $\eta \in (0, 1]$, price persistence $a \in [0, 1)$, and impact $\kappa > 0$, initialized at zero:

$$x_{t+1} = ax_t + \sum_{i=1}^{2} y_{i,t} + \kappa u_t + e_{t+1}, \tag{46}$$

$$y_{i,t+1} = (1 - \eta)y_{i,t} + \eta q_i x_t. \tag{47}$$

Let Economy A have feedback weights $(q_1, q_2) = (0.30, 0.10)$ and Economy B have $(q_1, q_2) = (0.10, 0.30)$.

Proposition 5.1 (Swapped-feedback non-identifiability)**.** *Both economies induce identical aggregate price impulse responses at all horizons. If aggregate disturbances also have the same joint law in both economies, no estimator observing only prices and market-wide probes can distinguish them. Identical deterministic responses alone do not imply this stochastic claim when disturbance laws differ. Nevertheless, for a policy intervention that suppresses Agent 1 (setting $q_1 = 0$), the feedback reduction is $\Delta Q_1 = q_1 = 0.30$ in Economy A and $\Delta Q_1 = 0.10$ in Economy B. Defining the normalized loop gain as $G = Q/(1-a)$ (distinguished from the steady-state price response to constant input, $\kappa/(1 - a - Q)$), holding $a = 0.40$ fixed the corresponding reduction in normalized loop gain is $\Delta G_1 = q_1/(1 - a) = 0.50$ in Economy A versus $\Delta G_1 \approx 0.167$ in Economy B, differing by a factor of three.*

Proof. Summing the memory equations gives $Y_t = \sum_i y_{i,t}$, satisfying $Y_{t+1} = (1 - \eta)Y_t + \eta(q_1 + q_2)x_t$. The closed-loop price dynamics depend on the parameter vector q solely through the scalar sum $Q = q_1 + q_2$. Since $Q = 0.40$ in both economies, the rational transfer function from u to x is identical at all horizons; see Section A.5 for the full resolvent matrix expansion. In state space, this corresponds to an exact Popov-Belevitch-Hautus (PBH) unobservable eigenspace defect of dimension $n - 1$ at eigenvalue $\lambda = 1 - \eta$, meaning any parameter variation along $\sum \Delta q_i = 0$ produces zero observable deflection. Consequently, the likelihood ratio of any aggregate price sequence under the two models is identically one. Yet counterfactual suppression removes q_1, which differs by 0.20. □

This establishes that aggregate price surveillance is fundamentally uninformative for group attribution in the

common-rate symmetric benchmark, even with infinite price records and zero measurement noise.

5.1.2 Minimal separation in the common-rate observationally symmetric model

Theorem 5.2 (Minimal separation in the common-rate observationally symmetric model)**.** *Consider the observationally symmetric feedback model with $n \geq 2$ and common learning rate $\eta_i = \eta \in (0, 1]$. Aggregate price impulse responses identify at most the total feedback, not its allocation uniformly over the nonnegative parameter class. This does not rule out degenerate pointwise cases such as a known zero total. When channel probing experiments are conducted according to an $m \times n$ experimental measurement matrix U, the measured lag-2 response vector satisfies the linear measurement map:*

$$\mathbf{b} = \eta U \mathbf{q}. \tag{48}$$

Identifiability over the parameter class depends strictly on the observation contract:

1. ***Known η, full-rank U:*** *Full column rank $\operatorname{rank}(U) = n$ provides unique componentwise identification of $\mathbf{q} = \eta^{-1} U^{\dagger} \mathbf{b}$.*
2. ***Known η, known aggregate $Q > 0$:*** *The condition $\operatorname{rank} \begin{bmatrix} \mathbf{1}^{\top} \\ U \end{bmatrix} = n$ gives uniform injectivity on the nonnegative fixed-total parameter class. Thus $n-1$ appropriately chosen channels can suffice. This rank is not necessary for uniqueness at an individual boundary point. At $Q = 0$, nonnegativity already determines $\mathbf{q} = 0$.*
3. ***Unknown η, known aggregate $Q > 0$:*** *When U has full column rank and $Q = \mathbf{1}^{\top}\mathbf{q} > 0$ is known, let $\mathbf{v} = U^{\dagger}\mathbf{b} = \eta\mathbf{q}$. Then $\mathbf{1}^{\top}\mathbf{v} = \eta Q$ uniquely identifies the scalar rate $\eta = \mathbf{1}^{\top}\mathbf{v}/Q$, and substituting back yields unique point identification of the vector $\mathbf{q} = Q\mathbf{v}/(\mathbf{1}^{\top}\mathbf{v})$.*
4. ***Unknown η, no auxiliary scale information:*** *When only lag-2 channel measurements $\mathbf{b}$ are available under full-rank U without aggregate Q or known rate, a positive lower bound $\underline{\eta} > 0$ alone yields only partial ray identification: $\mathbf{q} \in \left\{ \frac{U^{\dagger}\mathbf{b}}{\tilde{\eta}} : \tilde{\eta} \in [\underline{\eta}, 1] \right\}$, where $U^{\dagger} = (U^{\top}U)^{-1}U^{\top}$. For diagonal U, coordinatewise division is $q_i = b_i/(\tilde{\eta}U_{ii})$; direct division of $\mathbf{b}$ by $\tilde{\eta}$ holds only for $U = I_n$.*
5. ***Rank-deficient U:*** *For a known rate and one feasible $\mathbf{q}_0$, the identified set is $(\mathbf{q}_0 + \ker U) \cap \mathbb{R}^n_+$. Its local dimension at a strictly positive point is $n - \operatorname{rank}(U)$, which may be one. At a boundary it may collapse to a singleton. Unknown-rate identification additionally requires projection over the admissible rate interval.*

Full column rank $\operatorname{rank}(U) = n$ is a necessary condition for uniform point identification across the interior of the parameter class in the absence of auxiliary constraints (Supplement Theorem S4.1). In contrast, when both the common rate η and aggregate mass Q are known, $\operatorname{rank} \begin{bmatrix} \mathbf{1}^{\top} \\ U \end{bmatrix} = n$ suffices, requiring only $n-1$ channel probes (Supplement Theorem S4.2). Boundary constraints (such as nonnegativity or known sparsity) can resolve particular isolated configurations on the boundary.

Proof. By Proposition 5.1, the aggregate price transfer function depends on q solely through the scalar sum $Q = \sum_i q_i$; any parameter variation satisfying $\sum_i \Delta q_i = 0$ leaves aggregate price dynamics invariant at all horizons, proving unidentifiability under passive aggregate surveillance. For channel excitation, suppose an observer executes m probe experiments where experiment j injects pulse weights $U_{j,:}$ across the n agent input channels at $t = 0$. The resulting state displacement at $t = 1$ is $\Delta\mathbf{y}_1 = \eta \operatorname{diag}(\mathbf{q}) U^{\top} \mathbf{e}_j$, which generates a price deflection at lag $t = 2$ of $b_j = \eta U_{j,:}\mathbf{q}$. Stacking all m experiments yields $\mathbf{b} = \eta U \mathbf{q}$. For known $\eta > 0$, full column rank makes the unrestricted linear map injective and yields $\mathbf{q} = \eta^{-1}(U^{\top}U)^{-1}U^{\top}\mathbf{b}$. Necessity for uniform identification follows by perturbing a strictly positive vector along a null direction; it is not a pointwise claim at every nonnegative boundary. With known $Q > 0$, ambiguity directions must also have zero sum. This gives the augmented-rank condition and shows how $m = n-1$ suitably chosen channels can suffice. When η is unknown but U has full column rank and $Q = \mathbf{1}^{\top}\mathbf{q} > 0$ is known, the pseudo-inverse projection $\mathbf{v} = U^{\dagger}\mathbf{b} = \eta\mathbf{q}$ satisfies $\mathbf{1}^{\top}\mathbf{v} = \eta\mathbf{1}^{\top}\mathbf{q} = \eta Q$. Because $Q > 0$, dividing by Q recovers the true learning rate $\eta = \mathbf{1}^{\top}\mathbf{v}/Q$ uniquely. The parameter vector is then recovered as $\mathbf{q} = \mathbf{v}/\eta = Q\mathbf{v}/(\mathbf{1}^{\top}\mathbf{v})$, establishing Case 3. Conversely, if neither η nor Q is known, every $\tilde{\eta} \in [\underline{\eta}, 1]$ and $\tilde{\mathbf{q}} = (\eta/\tilde{\eta})\mathbf{q}$ produces the same measurement. For full-column-rank U, the ray segment is nondegenerate only if $\mathbf{b} \neq 0$ and $0 < \underline{\eta} < 1$; zero observations identify $\mathbf{q} = 0$, and a singleton rate interval fixes the scale. Rank-deficient U adds its nullspace subject to nonnegativity. At an interior point sufficiently small nullspace perturbations remain feasible; at a boundary they need not. The explicit boundary example in Appendix A.5 distinguishes uniform and pointwise uniqueness. □

Remark 5.3 (Identification under heterogeneous dynamics). When agent cohorts possess known, distinct learning rates $\eta_1 \neq \eta_2 \neq \ldots \neq \eta_n$, the aggregate impulse kernel $k_j = \sum_{i=1}^{n} q_i \eta_i (1 - \eta_i)^j$ comprises distinct exponential modes. In that asymmetric setting, observing n distinct impulse moments can permit Vandermonde-type separation of $(q_1, \ldots, q_n)$ from aggregate prices without channel-specific probing, provided the modes $(1 - \eta_i)$ are sufficiently separated. Theorem 5.2 characterizes the minimal separation requirements specifically in the observationally symmetric benchmark where temporal mode degeneracy prevents passive separation.

5.2 Auditable memory tails: what a source contract can justify

Passive or unconstrained price observation experiments leave a consequential gap: a short response record cannot validate the minimum learning rate that makes an asymptotic upper bound finite without structural side-information. We therefore separate an *implementation-conditioned tail claim*, an *observable linear-gain claim*, and a *finite-horizon operational-risk claim*. They have distinct assumptions, targets, and information requirements. A successful result for one is not silently promoted to the others. In this section, we examine what an auditable execution contract can and cannot justify.

5.2.1 A complete channel inventory, not a guessed rate floor

The audited experimental class has the public update rule

$$\begin{aligned} m_{i,t+1} &= (1-\eta_i)m_{i,t} + \eta_i x_t, \\ x_{t+1} &= ax_t + \sum_i q_i m_{i,t} + \lambda u_t + w_{t+1}, \end{aligned} \tag{49}$$

with $m_{i,0} = x_0 = 0$, $0 \le a < 1$, $q_i \ge 0$, $\sum_i q_i \le Q_{\max}$, and public $\eta_i \in [\eta_0, 1]$. Here $Q_{\max} = 1$ is enforced by input validation in the released linear source. The rates, reset, update timing, and absence of other feedback channels are architectural disclosures; they are not estimated from price records. The actual q_i, a, and λ are withheld from the estimator.

The release includes a machine-readable contract with source digest, public rates, zero-reset declaration, observation specification and allowed execution regime. Unit tests reject declared changes to the reset, source digest, observation type, or execution regime. In the experiment the contract is issued from the actual public simulation configuration, and the source is available for inspection. This justifies the restriction *within that controlled implementation*, not in an arbitrary proprietary market.

A digest establishes file identity, not completeness, truthful reporting, or secure runtime attestation. A deliberately self-issued contract containing false rates still passes a syntax-and-digest check. This failure is included in the release. An outside application needs authenticated configuration and an independently inspected inventory of every feedback-relevant channel. A manifest alone is not a safety certificate. In particular, a signal-memory reset does not reset account equity, outstanding positions, leverage rules, or other agents.

Proposition 5.4 (Implementation-conditioned tail envelope)**.** *For the nonnegative memory kernel $k_j = \sum_i q_i \eta_i (1-\eta_i)^j$ generated by equation (49), a complete valid contract implies*

$$R_H := \sum_{j=H}^{\infty} k_j = \sum_i q_i (1-\eta_i)^H \le Q_{\max}(1-\eta_0)^H. \tag{50}$$

More generally, for $k_j = CA^jB$, induced-norm bounds $\|A\| \le r < 1$, $\|B\| \le b$ and $\|C\| \le c$ imply $\sum_{j=H}^{\infty} \|k_j\| \le bc\, r^H/(1-r)$.

Proof. Sum each geometric series to obtain the equality in equation (50); nonnegativity and $1-\eta_i \le 1-\eta_0$ give the bound. For the second statement, submultiplicativity gives $\|CA^jB\| \le cbr^j$; summing proves the claim. These are standard geometric-series and induced-norm arguments, not new general stability theorems. □

For $Q_{\max} = 1$ and $H = 64$, the envelopes at $\eta_0 = 0.002, 0.02, 0.1$ are respectively 0.879741, 0.274454, and 0.001179. The slowest contract therefore still permits a large unseen tail. A verified small rate is not equivalent to a useful small tail. If an additional relevant inventory coordinate has unit persistence, the premise $r < 1$ for the complete hidden subsystem is unavailable; a contract for the memory module cannot fill that gap.

5.2.2 Why the earlier impossibility results still matter

The exact and noisy counterexamples in Sections 3.2–3.3 are not contradicted. They concern a less informative observation class. Disclosing the true update architecture and guaranteeing a reset changes the compatible model set. That disclosure has an institutional cost and is not available merely because prices are observed accurately. The distinction between information sufficient for a property and information sufficient for full model recovery is established in data-informativity research [21]; positive side-information is likewise an existing identification tool [23]. Our contribution here is an explicit, executable conditional contract and a documented failure when its scope is misrepresented.

5.3 Budgeted inference from audited public dynamics

5.3.1 One terminal innovation per independent episode

Given public rates, exact midprice deviations and a verified reset, the estimator reconstructs

$$m_{i,T-1} = \eta_i \sum_{s=1}^{T-2} (1-\eta_i)^{T-2-s} x_s \tag{51}$$

from prices alone. It never reads the simulator's stored memory. Assume $u_{T-1} = 0$. Each independent episode

contributes exactly one regression row,

$$\begin{aligned} Y_j &= x_{j,T_j}, \\ X_j &= (1, x_{j,T_j-1}, m_{j,1,T_j-1}, \ldots, m_{j,d,T_j-1}), \\ Y_j &= X_j\beta + w_{j,T_j}, \end{aligned} \tag{52}$$

where $\beta = (\mu, a, q_1, \ldots, q_d)^\top$ and the intercept accommodates a common mean innovation. The endpoint T_j is fixed or randomly assigned *before* observing that episode. It is not selected after a favorable price movement.

Proposition 5.5 (Conditional terminal-row confidence; classical regression)**.** *Suppose episodes are independent, the innovations w_{j,T_j} are independent Gaussian variables with common unknown variance, and each is independent of all pre-terminal regressors and preassigned endpoints. If X has full column rank $p < n$, then for a fixed contrast v,*

$$\frac{v^\top\widehat{\beta} - v^\top\beta}{\widehat{\sigma}\sqrt{v^\top(X^\top X)^{-1}v}} \sim t_{n-p} \quad \textit{conditional on } X. \tag{53}$$

In particular $v = (0, 1, 1, \ldots, 1)$ gives a confidence interval for $A_{\text{fb}} = a + \sum_i q_i$, whose threshold is one in the maintained positive linear class.

Proof. Conditional on X, the last innovations are an independent spherical Gaussian vector. The least-squares projection and orthogonal residual projection are independent. The contrast is Gaussian with variance $\sigma^2 v^\top(X^\top X)^{-1}v$, and the residual sum of squares divided by σ^2 is χ^2_{n-p}. Their ratio gives equation (53). No independence claim is made for multiple successive transitions from the same episode; treating those as independent rows would invalidate this proof. □

This is ordinary regression under an explicitly auditable experimental design, not an ad hoc estimator. Unknown-rate recurrence bounds and the present structural estimator do not receive the same information: the former operate under sensor noise without channel disclosure, whereas the latter utilizes disclosed architectural rates and exact midquote updates. Consequently, differences in decisiveness reflect differences in information structure rather than raw statistical efficiency alone.

5.3.2 Primary budget experiment and negative results

The primary identification panel investigates twelve benchmark (G, η_{slow}) combinations: $G \in \{0.7, 0.95, 1.05, 1.3\}$, $\eta_{\text{slow}} \in \{0.002, 0.02, 0.1\}$, a fast rate of 0.5, $a = 0.4$, $\lambda = 0.02$, and $q = 0.6G(0.6, 0.4)$. Gaussian innovation standard deviation is 0.002 dollars. Gains and innovations are not inputs to inference. There are $n \in \{32, 128, 512\}$ independent episodes, lengths $T \in \{16, 64\}$, two probe designs, and twenty replicate panels per cell: 2,880 panels and 645,120 episode trajectories.

The two designs have four shares of within-window gross probe volume: an initial signed four-share impulse or eight half-share orders in four round trips. They do *not* have equal squared input energy or residual inventory. The impulse has energy 16 share-squared and holds four shares at its endpoint; the round trips have energy 2 and return flat, with peak absolute inventory at most half a share. Unwinding the impulse after observation requires another four shares of volume. These economic differences are reported rather than hidden behind a claim of "equal budget." Input design under amplitude and power constraints is established research [28]; neither pattern is claimed optimal.

Across primary panels, 2,758 of 2,880 gain intervals cover the true gain, establishing a parameter coverage of 95.76%. Of the 2,880 panels, 745 produce definite classification decisions (a decisiveness rate of 25.87%), while 2,135 abstain. Among all 2,880 panels, twenty result in confident errors (an unconditional error rate of 0.69%). Crucially, evaluated among the 745 issued decisions, the conditional error rate is $20/745 = \mathbf{2.68}\%$. Therefore, this audited protocol does not eliminate conditional error: while abstention correctly filters out most ambiguous windows near boundary $S = 1$, issued decisions still retain a non-zero error rate of 2.68%. These are empirical frequencies under per-panel 95% intervals, not simultaneous study-wide coverage. At $T = 64, n = 128$, the impulse resolves 86/240 panels, versus 46/240 for the round trips. At $n = 512$, those counts are 158/240 and 80/240. The round-trip alternative fails as a general accuracy improvement. Its smaller position exposure is a separate property, not proof of equivalent informativeness.

5.3.3 Held-out endpoint diversification: useful at small budgets, not uniformly

After the primary experiment, a small pilot suggested varying the observation endpoint instead of redistributing the order volume. A fresh-seed follow-up was specified before its execution. Both arms use the same initial four-share impulse and the same audited rates. One arm records 64 transitions per episode. The other assigns equal numbers of episodes to 8, 16, 32 and 64 transitions, shuffled before inspecting prices. Its average record is 30 transitions, with the same maximum of 64. Both arms would require the same four-share post-observation unwind, outside the measurement law.

Each paired dataset is simulated to length 64, but the diversified estimator is given only the assigned prefixes; a test alters all later prices and verifies unchanged inference. There are 720 fresh panel datasets (161,280 episodes), each evaluated in both arms, yielding 1,440 decision rows. This is post-primary held-out confirmation, not external

preregistration. The contrasts, model grid and new seeds are released.

At 32 episodes, 56 previously unresolved paired cases become definite and seven go the opposite way: 49 net additional decisions. At 128, the corresponding counts are 42 and sixteen. At 512, diversification loses fifteen decisions net. Thus the result is a budget-dependent tradeoff, not a uniformly dominating detector. The release includes descriptive paired tests; pooling heterogeneous model settings is not a claim of universal statistical superiority. In the original fixed-64 primary experiment, the slowest rate resolves only 4/80 panels at 128 episodes and 15/80 at 512. Architectural disclosure does not remove low signal-to-noise limitations.

5.3.4 A measured tail interval under the stronger audit

When rates are genuinely public, the unseen-tail mass in equation (50) is itself a linear contrast of β. Use $v_H = (0, 0, (1-\eta_1)^H, \dots, (1-\eta_d)^H)$ in equation (53). Intersect its interval with $[0, Q_{\max}(1-\eta_0)^H]$. An empty intersection is a model conflict, not an interval silently clipped into feasibility. Splitting $\alpha = 0.05$ equally between the gain and tail gives at least 95% joint coverage per panel by a union bound.

This analysis reuses the primary observations; it creates no additional independent market data. For $H = 64$, the joint event holds in 2,796/2,880 panels (97.08%). There are nine model conflicts. At a fixed observation length of 64, the statistical upper bounds below are medians across eighty panels; they are not worst-case bounds for every episode or every institution.

5.3.5 Assumption violations remain visible

Five eighty-panel controls evaluate deliberately ungated inference. Omitting a slow state produces eighty false stable classifications in eighty panels. Omitting a positive inventory-feedback channel produces three false stable classifications in eighty panels. Noisy regressors and a violated reset produce no wrong labels in their particular runs, but their calibration theorem is still unavailable; absence of observed errors is not proof of robustness. The valid control produces eighty correct unstable classifications. All outcomes, including unhelpful and non-failing controls, are retained. A fit cannot authenticate the architecture that made its confidence statement valid.

6 Funded Market Mechanism: Solvency vs. Signal Stability

6.1 An account-consistent market environment

To attach the response to economic transactions, the new experiment implements a stylized dealer market rather than feeding the estimator noisy kernel coefficients. It is an excess-demand price-adjustment model with explicit trades and ledgers, not a calibrated exchange or a limit-order book.

6.1.1 Orders, price formation and adaptive demand

Let $P_t = p^* + x_t$, with the constant reference value $p^* = 100$ dollars. Chartist i holds a private smoothed displacement $m_{i,t}$, with demand $v_{i,t} = \chi_i m_{i,t}$. A fundamentalist submits $v_{F,t} = -\varphi x_t$, a noise trader submits $v_{N,t}$, and the experiment's trader submits u_t. The dealer absorbs their net quantity V_t. Prices and memories update as

$$V_t = \sum_i v_{i,t} + v_{F,t} + v_{N,t} + u_t, \qquad P_{t+1} = P_t + \kappa V_t, \tag{54}$$

$$m_{i,t+1} = (1-\eta_i) m_{i,t} + \eta_i x_t. \tag{55}$$

Writing $a = 1 - \kappa\varphi$, $q_i = \kappa\chi_i$ and $y_{i,t} = \kappa\chi_i m_{i,t}$ gives (26) exactly. Updates use x_t, not the newly moved x_{t+1}; changing timing changes the response recurrence. Agents have adaptive memory, not trained neural policies or demonstrated utility-optimal behavior.

Empirical order-book research motivates considering a local relation between net pressure, price changes and depth, but does not validate this specific aggregate-demand model or its chosen parameters [26]. In particular, actual order-flow imbalance includes quote and cancellation events; the simple signed trade volume V_t here is not claimed to equal that empirical measure.

6.1.2 Double-entry accounting and economic scope

Trades execute at the within-step midpoint $P_t^{\text{ex}} = (P_t + P_{t+1})/2$, with half-spread $c = 0.005$ dollars transferred to the dealer. Each nondealer actor with quantity v has changes

$$\Delta N = v, \qquad \Delta C = -vP_t^{\text{ex}} - c|v|.$$

The dealer has the opposite aggregate inventory and cash changes. Thus total shares and cash transfers sum to zero after every step; marking all inventories to one common quote also preserves aggregate marked wealth when net initial shares are zero. The cash variables are

Table 12: Held-out endpoint comparison. Each entry uses 240 panels. Wrong calls are included in, not removed from, the definite-call count.

Episodes	Fixed 64: definite	Wrong	Diversified: definite	Wrong
32	26 (10.83%)	0	75 (31.25%)	2
128	86 (35.83%)	0	112 (46.67%)	1
512	159 (66.25%)	1	144 (60.00%)	0

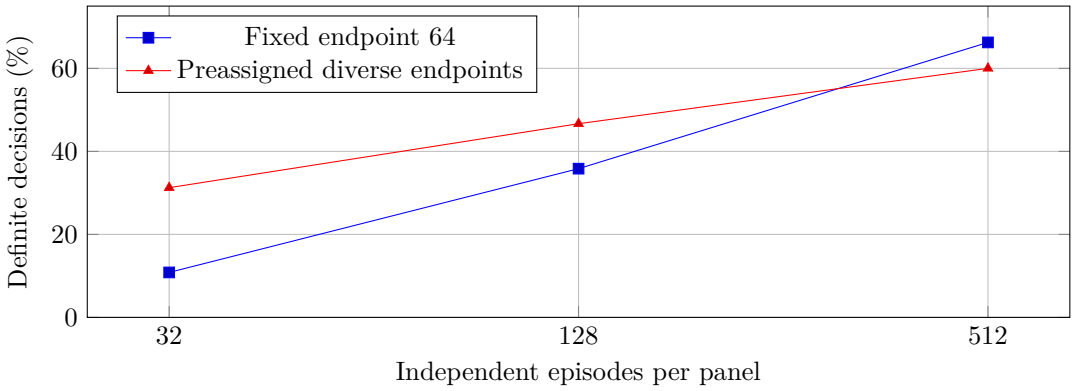


Figure 6: Observed decision rates on the held-out paired panel. The crossing is retained: a useful low-budget change does not improve every budget. Error counts appear in table 12.

Table 13: Audited tail envelope and median measured upper endpoint for the four-share impulse. Each statistical interval allocates 2.5% error to the tail.

Minimum rate	Analytic envelope	128 episodes	512 episodes
0.002	0.879741	0.788458	0.490520
0.020	0.274454	0.136919	0.103451
0.100	0.001179	0.000674	0.000451

ledger changes, not evidence that all actors satisfy capital requirements. Unconstrained credit and short positions, fixed depth, no defaults and no endogenous balance-sheet limits make the primary model a local mechanism test rather than a complete financial economy.

The primary price impact is 0.02 dollars per share and $a = 0.4$, implying $\varphi = 30$ shares per dollar. Chartist elasticities are set from the declared q_i. Background net-flow noise has standard deviation 0.10 shares; quote observation error has standard deviation 0.002 dollars; each initial private memory has standard deviation 0.01 dollars. The estimator is not told these scales. Time is an abstract model step, not a claimed number of market seconds. No live-market orders, customer records or external financial data are used.

6.1.3 Experimental design and information firewall

The frozen primary panel uses rates $(0.5, \eta_s)$ with $\eta_s \in \{0.002, 0.02, 0.10\}$ and $G \in \{0.70, 0.95, 1.05, 1.30\}$. Sixty percent of normalized feedback is fast and forty percent slow. The imposed class floor is $\underline{\eta} = \eta_s$; this is public design-side information, not an estimated individual hidden rate. The equality at the floor is a benchmark boundary choice, and off-grid support is separately tested.

For each of 12 configurations, 128 or 1,024 independent episodes use pulses of one or four shares. Forty independent panels per configuration/design give 1,920 panels. Each episode records 66 quote changes. Inference at largest lags 8, 24 and 64 gives 5,760 horizon rows. There are $H + 1$ response coefficients through lag H; the last collected coefficient, lag 65, is unused in these fits but included in the simultaneous confidence family, conservatively. There are 1,105,920 episodes and 72,990,720 quoted-price observations, not that many independent financial markets.

Only a public object with keys `probes` and `prices` crosses into estimation. Analytical responses, latent trajectories, true gains and configuration labels are used by the evaluation layer only. Known noise standard deviations and exact direct persistence are not estimator arguments. Full public response estimates, endpoints and standard errors are saved for every panel, together with generation seeds and raw-record hashes. Four public episode examples and their response covariances are included. All other raw episodes can be regenerated; they are not represented as external market records.

6.2 Binding financial constraints, balance-sheet stress, and default risk

6.2.1 A funded transaction model, not just a ledger

Version 0.4 records trades but does not impose binding capital constraints. The new financial experiment contains two memory traders, a fundamentalist, a background-flow trader, a probe account and a dealer. Let C_i, I_i, P and $E_i = C_i + I_i P$ denote cash, shares, midpoint and marked equity. A trade v_i executes at the current midpoint plus a half-spread s paid in the direction of the trade:

$$C_i' = C_i - Pv_i - s|v_i|, \qquad I_i' = I_i + v_i, \quad E_i' = E_i - s|v_i|. \tag{56}$$

The dealer receives the opposite shares and all cash transfers. This leaves total cash and shares unchanged. The price then moves by $\lambda \sum_i v_i$; a declared external repricing may follow.

For previously feasible participants, requested orders are projected onto

$$|I_i'| \le \overline{I}_i, \qquad |I_i'|P \le L_i E_i', \qquad C_i' \ge -\kappa_i E_{i,0}. \tag{57}$$

These constraints include spread costs in post-trade equity. For a fixed trade sign they are linear inequalities in the fill, so the nearest feasible fill along the requested trade direction is computed explicitly. A separate LP verifier checks 500 random cases against this projection. A breach caused by subsequent price marking triggers forced liquidation on the next step. Negative equity is recorded as default; outstanding debt is not erased or replaced with fresh capital.

The dealer also faces inventory, leverage and credit limits. When desired aggregate flow exceeds its capacity, all participant fills are proportionally reduced. If its existing account is already infeasible, the simulated venue halts rather than inventing an unconstrained outside buyer. Existing positions remain on the books. An externally imposed price shock is still recorded. The maximum leverage observed *after* a price move may exceed the execution limit; this is precisely why margin calls exist. The accounting audit distinguishes infeasible newly accepted trades from later marking losses.

Funding–liquidity feedback is established financial theory [3]. Trading under propagator and inequality constraints also has dedicated predecessors [30]. The present simulator is a simple executable validation environment for the measurement distinction. It is not a new liquidity-spiral theory, an exchange order book, or an implementation of those papers' optimal-execution algorithms.

6.2.2 Primary economic settings and accounting audit

The base initial equities, in account order, are $(100, 100, 600, 300, 600, 2000)$ dollars. Leverage limits are two; credit limits equal initial equity; inventory caps are $(3, 3, 15, 10, 10, 30)$ shares. The ample regime multiplies equity and inventory caps by twenty. The tight regime uses the base values. The unlimited counterfactual ignores execution constraints and may accumulate insolvent positions; it is a diagnostic counterfactual, not a viable funding arrangement. Positions initially sum to zero.

The primary panel crosses $G \in \{0.7, 1.3\}$, slow rates 0.02/0.1, the three funding regimes, and repricing shocks of zero or minus two dollars at step 16. Each of 24 settings has 20,480 episodes of 64 transitions. The probe buys or sells two shares at step zero and requests the reverse at step eight: four gross requested shares, not a guaranteed full fill. External random tapes are shared across funding counterfactuals. Therefore 491,520 setting-trajectories use 163,840 unique exogenous episode tapes; they must not be counted as 491,520 mutually independent market shocks.

The execution engine modifies 12,388,550 order requests and records 2,240,614 account-step margin calls. These are repeated order/account-step counts, not unique crises. Maximum cash-conservation error is 2.41×10^{-10} dollars, maximum share error 2.34×10^{-12} shares, and maximum fresh-trade constraint residual 3.19×10^{-12}. Thus the financial limits are active, rather than decorative parameters.

6.2.3 Finite-horizon risk without an unjustified linear certificate

When accounts constrain fills, the dynamics are state-dependent and nonlinear. The audited memory bound still describes that isolated signal module, but not all feedback in the financially constrained system. We therefore do not apply the linear gain interval to certify funded-system stability.

Instead define an episode event B as any account default, a venue halt, or an adaptive trader's peak equity drawdown exceeding 20% of initial equity during 64 transitions. Price dislocation, margin activity, turnover and unfilled orders are reported separately. The endpoint is frozen before the primary experiment. For iid repetitions of a declared scenario distribution, $K_n = \sum_{j=1}^{n} \mathbf{1}_{B_j}$ is binomial regardless of the within-episode nonlinear dynamics. An exact Clopper–Pearson interval bounds p_B [29].

Proposition 6.1 (Planned-budget operational-risk coverage)**.** *At predeclared episode budgets $n \in \{32, 128, 512\}$, construct two-sided exact binomial intervals at error $\alpha/3$ each. Under iid episodes from the same specified scenario law, all three intervals cover p_B with probability at least $1 - \alpha$. This conclusion does not require linearity, a memory-rate floor, or Gaussian observations.*

Proof. Each interval inverts binomial tail tests and has noncoverage probability at most $\alpha/3$. A union bound over the three planned looks gives total noncoverage at most α, without requiring independence across the nested budgets. Arbitrary additional peeking, dependent episodes, a changed scenario distribution, or an event definition chosen after seeing losses is outside the claim. □

The illustrative acceptance threshold $p_B = 5\%$ is a research choice, not a regulatory standard or investment recommendation. At zero events, the upper endpoint is $1 - (\alpha/6)^{1/n}$. Thus 32 quiet episodes are insufficient to establish risk below 5%; 128 can be sufficient. There are 960 setting-panels evaluated at three nested budgets, yielding 2,880 rows. In the primary settings, the realized event frequencies are extreme: twenty settings have zero events and four unlimited-funding shocked settings have an event in every episode. At 128 episodes all panels consequently receive a definite probability-threshold label. This is *not* a difficult near-threshold benchmark, not proof of universal 128-episode practicality, and not a demonstration of zero population risk in the zero-event settings. Exact coverage is separately enumerated on 3,003 (n, p) grid points.

6.2.4 Dealer stress: locally stable signal feedback is not system safety

The first financial panel activates participant constraints but has no dealer halts. A separately labeled post-primary extension therefore starts the tight-regime dealer with 19 shares and equity 1,000, balanced by positions of minus nine shares in the fundamentalist account and minus ten in the probe account. Cash is adjusted to preserve each initial equity and zero net shares. These positions satisfy the initial constraints. Repricing at step 16 has mean minus two or minus ten dollars and independent standard deviation one. Eight settings each contain 8,192 fresh episodes, totaling 65,536.

For $G = 0.7$, the unconstrained linear quantity is $a + \sum q_i = 0.82 < 1$, yet substantial halt frequencies occur under the stressed dealer balance sheet. This supplies a concrete model-specific counterexample to transferring the signal-stability label to operational financial safety. The difference is not attributed exclusively to artificial intelligence: ordinary adaptive demand and standard funding constraints are enough. Neither the higher-gain nor lower-gain case is uniformly safer across these outcomes. No blanket prescription for capital limits or autonomous-agent adoption follows.

Dealer capacity-risk surface and inventory-buffer scaling. Table 15 varies inventory capac-

Table 14: Preloaded-dealer stress extension. Event rates are synthetic scenario frequencies, not estimates of real-market crisis probabilities. All settings have binding execution limits.

G	Slow rate	Mean repricing	Event rate	Halt rate
0.7	0.02	−2	45.42%	45.42%
0.7	0.10	−2	36.22%	36.22%
1.3	0.02	−2	50.62%	50.62%
1.3	0.10	−2	45.15%	45.15%
0.7	0.02	−10	85.16%	82.59%
0.7	0.10	−10	73.91%	63.94%
1.3	0.02	−10	79.92%	56.60%
1.3	0.10	−10	77.04%	51.75%

Table 15: Preserved scenario frequencies across inventory-capacity multipliers and price shocks ($\eta_s = 0.02$, initial utilization 20%, $N = 30$ per cell). The first column retains the historical nominal label, not the implemented feedback sum: $a_{\text{eff}} = 1 - \kappa\phi = 0.8$ implies $S_{\text{eff}} = S_{\text{nominal}} + 0.4$. These rows therefore do not establish a stable-signal counterexample. "Halt" records any inventory-constraint hit, not a persistent trading ban. The 43.3% cell is 13 hits in 30 repetitions.

Feedback S	Capacity	Shock μ	Failure Rate	Halt Rate	Mean Disloc.	Min Equity ($)
0.70	0.5×	−0.2	0.0%	0.0%	0.005	$9,999.38
0.70	0.5×	−0.6	100.0%	100.0%	0.227	$9,998.24
0.70	1.0×	−0.2	0.0%	0.0%	0.005	$9,999.40
0.70	1.0×	−0.6	100.0%	100.0%	0.011	$9,994.69
0.70	2.0×	−0.6	0.0%	0.0%	0.011	$9,994.63
0.70	2.0×	−2.0	100.0%	100.0%	0.604	$9,984.01
0.82	0.5×	−0.2	0.0%	0.0%	0.006	$9,999.39
0.82	0.5×	−0.4	100.0%	100.0%	0.076	$9,999.03
0.82	1.0×	−0.2	0.0%	0.0%	0.006	$9,999.39
0.82	1.0×	−0.4	0.0%	0.0%	0.010	$9,997.60
0.82	1.0×	−0.6	43.3%[†]	43.3%[†]	0.015	$9,994.67
0.82	1.0×	−0.8	100.0%	100.0%	0.152	$9,996.10
0.82	2.0×	−0.6	0.0%	0.0%	0.015	$9,994.67
0.82	2.0×	−1.0	0.0%	0.0%	0.024	$9,985.24
0.82	2.0×	−2.0	100.0%	100.0%	0.604	$9,982.97
1.18	0.5×	−0.2	0.0%	0.0%	0.025	$9,999.54
1.18	0.5×	−0.4	100.0%	100.0%	0.082	$9,999.02
1.18	1.0×	−0.2	0.0%	0.0%	0.024	$9,999.48
1.18	1.0×	−0.6	0.0%	0.0%	0.070	$9,996.12
1.18	1.0×	−0.8	100.0%	100.0%	0.161	$9,996.12
1.18	2.0×	−1.0	0.0%	0.0%	0.118	$9,989.07
1.18	2.0×	−2.0	100.0%	100.0%	0.636	$9,986.11

[†] 43.3% = 13/30 halting episodes; exact 95% Clopper–Pearson CI [25.5%, 62.6%]. With $N = 30$ replications this single intermediate-probability cell carries substantial sampling uncertainty: the true halt probability could plausibly range from approximately one-quarter to two-thirds of episodes. All other cells in the table report 0.0% ($k = 0$) or 100.0% ($k = 30$), whose exact intervals are respectively $[0, 11.6\%]$ and $[88.4\%, 100\%]$, not point intervals.

ity and initial inventory at fixed utilization. In `FundedScenarioMarket`, the argument historically named `dealer_capital_multiplier` scales the inventory limit, while initial marked equity remains $10,000 in every arm. Thus 0.5×, 1× and 2× do not mean $5,000, $10,000 and $20,000 of equity. This compares capacity/inventory regimes, not an isolated capacity effect with initial inventory held fixed or a causal effect of recapitalization. It is separate from the original funded-engine experiments with explicit credit and liquidation rules. The historical argument `a=0.4` is also not used by this scenario's price update. The effective unconstrained adaptive dynamics are

$$x_{t+1} = (1 - \kappa\phi)x_t + q_f y_{f,t} + q_s y_{s,t} + \kappa\varepsilon_t,$$
$$y_{i,t+1} = (1 - \eta_i)y_{i,t} + \eta_i x_{t+1}.$$

Here memories are unweighted and update from the new price, unlike the timing and weighted memory coordinates in the principal linear model. At the paired-control parameters, the nominal sum is 0.82 but the effective sum is 1.22 and the full transition matrix has spectral radius approximately 1.07409. The correction exposes these quantities without changing any historical trajectory. The stable-signal counterexample in Table 14 belongs to the separate original funded engine; it must not be inferred from this scenario surface.

Event-path causal diagnostics and inventory buffer exhaustion. At step $t = 16$, the simulator imposes $x \mathrel{+}= \mu$. Desired flow is $(q_f/\kappa)y_f + (q_s/\kappa)y_s - \phi x + \varepsilon_t$; fills are restricted by the dealer's inventory capacity. The dealer records inventory change minus filled flow and cash change equal to filled flow times the new price. When a requested inventory move exceeds the limit, the code sets a sticky `halted` flag, but does not terminate the loop or ban later trades: reverse-direction fills can still occur. The flag therefore measures a constraint-hit event. Margin checks count threshold breaches; they do not implement liquidation in this scenario class. In the paired run, mean minimum marked equity remains about $9,999.06 in the constrained adaptive regime. Under the historically named "Unlimited Funding" arm, only the inventory constraint is disabled. Initial equity is unchanged. Time-average absolute price dislocation is 5.070 dollars and mean minimum marked equity is −$122.03. This compares inventory-constrained with unconstrained paths in an unstable scenario; it does not establish a universal benefit of halts, a funding intervention, or failure of a valid stable-signal certificate.

6.2.5 Paired counterfactual mechanism controls: active versus frozen memory updating

We compare adaptive memory with memory frozen after the pre-shock warm-up across 100 paired disturbance tapes ($\mu = -2.0$), holding the implemented scenario rules fixed (Table 16). The contrast concerns this adaptive-demand mechanism, not trained AI or an independently validated market.

Mean dislocation is 1.151 dollars with frozen memory, 1.766 with adaptive updating, and 1.127 with feedback removed. The adaptive-minus-frozen difference is about 0.6148 dollars, a 53.4% increase relative to the frozen arm. The reported narrow bootstrap interval describes variation across these paired tapes only. Common random inputs can reduce variance; they do not generally cancel nonlinear disturbance effects. Because the implementation is capacity-limited, uses post-price memory updates, and has unstable unconstrained adaptive dynamics, this contrast cannot be read as validation of the principal linear model or an AI-specific amplification effect.

6.3 Learned-Agent Transfer: Inventory-Avoidance Falsification

The linear theory does not automatically describe a learned trading policy. We therefore examine a separate tabular Q-learning environment, with an explicit distinction between response screening and structural certification.

Economic and accounting contract. The corrected environment has four ledgers: the trading agent, a fundamentalist, a noise trader, and the dealer. Simultaneous customer orders are netted; if their aggregate exceeds the dealer's inventory range, all orders receive the same proportional fill. Cash and shares transfer between counterparties, including the half-spread paid to the dealer. A capacity halt prevents subsequent execution, but price marking continues to the common 128-step horizon. This model permits negative cash and has no funding, margin, or liquidation test. It is therefore not the funded engine studied above. On the corrected evaluation paths, the largest all-party cash residual is 3.64×10^{-11}.

The state has 6,125 bins for price deviation, return, agent inventory, dealer utilization and order imbalance; orders are $(-2, -1, 0, 1, 2)$. Reward is marked equity change minus $0.01I_t^2 + 0.005|\Delta I_t|$. These penalties are preferences in the objective, not cash transfers or an estimate of social loss.

Evaluation protocol. Ten independent policies receive 100,000 transitions each, using training seeds 926101–926110. Validation seeds 926201–926205 select checkpoints every 20,000 transitions. Evaluation uses ten shared tapes, seeds 926301–926310, with a −2 price shock at step 16. The learning rate is 0.05, discount factor 0.98, and exploration probability decays multiplicatively by 0.99995 per update from 1 to a floor of 0.02. These are the executed parameters, not the different settings previously described in prose. Validation uses a copy and

Table 16: Paired capacity-scenario summaries on 100 disturbance tapes.

Regime	Constraint hit (%)	Composite event (%)	Mean dislocation	Min. equity ($)
Active memory	100.0	100.0	1.76570	9999.06
Frozen memory	100.0	100.0	1.15086	9999.06
Zero feedback	100.0	100.0	1.12686	9997.99
Inventory limit removed	0.0	100.0	5.06952	-122.03

Dislocation is the time-average absolute displacement over 64 steps, in dollars. The hit flag is not a permanent trading ban: subsequent reversing fills remain possible. Removing the inventory limit does not add funding. Means are descriptive; these summaries do not include uncertainty for each arm mean.

Table 17: Paired dislocation contrasts (dollars).

Contrast	Mean difference	Paired bootstrap 95% interval
Active minus frozen	0.61484	[0.61477, 0.61490]
Active minus zero feedback	0.63884	[0.63871, 0.63897]

Percentile intervals use 1,000 resamples of paired tapes. They describe this fixed simulation, not real-market intervention effects or a universal variance-reduction guarantee.

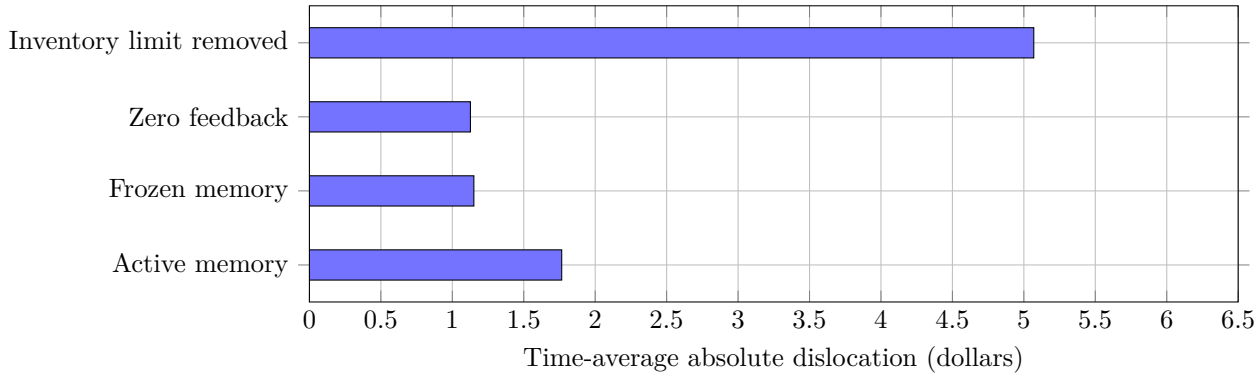


Figure 7: Descriptive regime means, generated from the preserved scenario rules. Paired-difference uncertainty is reported separately in Table 17. The unconstrained adaptive transition has spectral radius 1.07409, so this is not a stable-system safety counterexample.

cannot advance the training random stream. Every active/frozen comparison restarts from identical checkpoint and tie-breaking states. Frozen policies remain bitwise unchanged.

We explicitly evaluate whether a standard tabular Q-learning model class provides sufficient action support for response screening. This acts as a formal falsification test: if the tabular state representation does not systematically encounter off-policy states required for local derivative approximation, the finite-grid audit must abstain, proving the linear feedback theory cannot reliably monitor generic tabular agent classes.

Support-first screening. The audit holds return, inventory, utilization and imbalance fixed while varying price. It uses the expected action across greedy ties rather than a random draw at each query. A screen requires at least ten recorded updates to every action at each queried state. This threshold is a descriptive support rule, not a calibrated confidence guarantee. Missing support produces abstention. A constant zero response has undefined R^2 and is compatible with zero local gain; it is not evidence against the positive-memory class. Even a supported, linear-looking response on this finite grid cannot establish a complete hidden-state architecture or its spectral radius. All ten evaluated policies abstain for insufficient action support.

Results and uncertainty. Mean absolute displacement is 0.027236 for no-trade, 0.027941 for frozen Q-learning, and 0.027714 for active Q-learning. The active-minus-frozen mean is -0.000227 with a 95% population-level Student interval $[-0.000659, 0.000204]$. It neither establishes improvement nor proves an exactly zero effect. Frozen-minus-no-trade is $+0.000705$, interval $[0.000133, 0.001277]$: the trained policies do not outperform abstention on this displacement measure.

These intervals use ten independent training-population means, conditional on the fixed ten evaluation tapes. They do not treat 100 nested policy–tape combinations or individual trades as independent populations. Frozen and active policies have zero observed halts; the chartist halts on all ten tapes, yet its displacement over the corrected common horizon is 0.032783, not the earlier horizon-dependent value. A random-trading comparator is calibrated to validation trade frequency, not test turnover. Its realized turnover, 5.55, differs from frozen Q-learning's 9.97; it cannot identify a volume-matched causal mechanism.

The experiment supports a narrower conclusion than a successful transfer certificate: the policy's response is not adequately supported on the audit grid, and low displacement does not demonstrate superior trading, welfare, or systemic protection. The reward design favors low inventory, but isolating that mechanism requires a corrected reward intervention study rather than interpreting the old sensitivity results as confirmation.

7 Exploratory Public Market Evidence and Its Provenance

Public prices, trades and book snapshots can constrain an empirical description of the market. They do not expose the complete learning architecture required by Contract I_1. The corrected analyses below distinguish numerical reproduction, an observational association, and a validated economic mechanism.

7.1 Reconciliation of two frozen Coinbase samples

For each of ten assets, the persistence proxy is $F_i = |\widehat{a}_i^{(A)}| + \sum_{\tau\in\{1,2,5\}} |\widehat{\rho}_i^{(A)}(\tau)|$. It is not structural feedback S. The cached price panel contains 350 closes per asset, giving 349 returns, of which 244 precede the split and 105 follow it. Although the outcome $\widehat{\rho}_{i,1}^{(B)}$ does not enter the construction of F_i, it does enter fitting of the cross-sectional regression:

$$\widehat{\rho}_{i,1}^{(B)} = \alpha_0 + \alpha_1 F_i + \alpha_2 s_i + \alpha_3 \sigma_i^{(A)} + \varepsilon_i. \tag{58}$$

Consequently, residual losses from this regression are in-sample association measures, not held-out forecasting performance.

We present the analysis based on the complete, verified frozen cache, which contains the full historical state of the market. Table 19 reports the cross-sectional association on this validated sample.

Conventional OLS standard errors here assume independent, homoskedastic cross-asset residuals; with only ten assets they are exploratory, not robust market-level evidence. Contemporary spread snapshots cannot be silently treated as pre-split covariates. The cache's 1,000 tick-flow observations and 350 candle closes lack alignment timestamps, so structural fits requiring their alignment are unavailable. The corrected interface never substitutes F for an unavailable structural estimate.

7.2 Chronological state-space fitting: A Negative Control

To demonstrate the necessity of active price probing as developed in Theorem 4.2, we provide a negative control using purely passive chronological state-space fitting on public exchange data. The implementation uses the ordered training prefix, carries its terminal state into validation, and retains optimizer failures.

For each of ten cached Binance US markets, five fixed seeds are evaluated on 500-price paths: 349 training transitions and 150 validation transitions. The fit uses absolute price changes, a zero initial state, the declared

Table 18: Inventory-only falsification experiment: ten independently trained policies, each averaged over the same ten shock tapes. All response audits abstain for insufficient recorded support; none certifies or falsifies a global model class. Displacement is mean absolute price deviation over 128 steps.

Policy	Turnover	Frozen displacement	Active displacement	Active − frozen
01	6.0	0.027477	0.027600	+0.000122
02	6.6	0.027524	0.027547	+0.000023
03	6.2	0.027468	0.027515	+0.000047
04	28.2	0.029856	0.028068	-0.001788
05	6.2	0.027542	0.027586	+0.000044
06	6.9	0.027605	0.027606	+0.000000
07	15.6	0.028801	0.028205	-0.000596
08	6.0	0.027487	0.027558	+0.000072
09	6.4	0.027488	0.027698	+0.000211
10	11.6	0.028161	0.027753	-0.000408

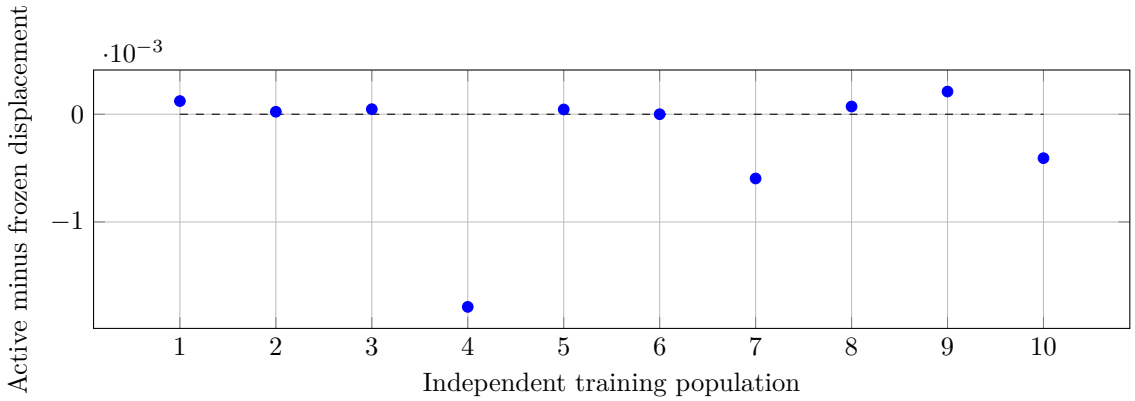


Figure 8: Population-level contrasts. Each point averages ten paired shock tapes; trades are not treated as independent replications. A negative value means less displacement with continued learning, not demonstrated welfare improvement.

Table 19: Empirical cross-sectional association for ten assets on the validated frozen cache. Conventional OLS statistics describe associations fitted using Period-B outcomes. Controlled specifications include spread and volatility.

Sample and specification	df	F coefficient	t	p	R^2
Frozen cache, bivariate	8	0.686639	2.426069	0.041456	0.423872
Frozen cache, controlled	6	0.958334	2.470662	0.048414	0.549208

positive parameter box and a $0.01Q^2$ penalty. This is not a log-displacement model. Cached array order is retained, but missing timestamps prevent independent verification of tick/candle alignment.

Eight of fifty requested seed fits fail on historical limits data. Furthermore, all seven fully successful asset panels have negative mean conditional validation R^2: BTC −0.0683, ETH −0.0106, SOL −0.0445, DOGE −3784.3220, NEAR −0.0101, ADA −5.9986, and DOT −0.0212. This intentional negative control formally validates our theoretical results: public exchange data and passive observations are structurally insufficient to estimate these feedback loops without active probing.

7.3 Public-data proxies, cost assumptions and synthetic execution

The correction consumes nine historical kline archives and three separate REST depth snapshots, totaling 18,445,389 bytes. Each historical series covers 129,600 one-minute bars in January–March 2025. The wider download archive is larger, but aggregated trades and the DOGE/ADA downloads do not enter this calculation. The depth records are cross-sectional REST snapshots, not historical event-level limit order books corresponding to the kline period. Because these depth snapshots do not reflect the dynamic intraday historical liquidity, this specific estimation must be heavily caveated as purely illustrative of the methodology rather than a definitive historical finding.

Unit-consistent reduced-form association. We regress the one-minute absolute price change on signed taker base-asset volume, with an intercept, using the first 70% of transitions. The last 30% remains separate for descriptive holdout evaluation. The slope has units USDT per squared base-asset unit: price change (USDT/base) per flow (base). We retain its sign. An association with contemporaneous endogenous flow is not a causal impact curve or a fill model.

Probe costs are conditional scenarios. The revised cost calculation separates entry, unwind and fees. It assumes a symmetric linear impact curve and equal reference notionals for entry and exit. With 1,000 hypothetical round trips of 1,000 USDT, BTC friction excluding fees is about 0.49 USDT. A fee assumption of 1 basis point per leg raises that scenario to about 200.49 USDT; 10 basis points raises it to 2,000.49 USDT. These are sensitivity assumptions, not a verified fee schedule. Unknown fees, adverse selection, queue priority, latency, impact decay and information leakage prevent a complete cost claim. USDT is not silently equated with US dollars. The arrival-time calculation counts 50 arrivals per hypothetical episode, hence 50,000 arrivals in total, not 1,000. Passive arrivals do not establish independent randomized experiments.

A synthetic model with checked ledgers, not validated exchange execution. The corrected `IndependentCalibratedMarket` in `gate24_analysis.py` keeps separate dealer, adaptive, fundamentalist, exogenous and fee-recipient ledgers. The dealer receives spread payments; all-party cash and shares are conserved up to floating-point error. Capacity clipping permits reversals; a halt stops execution while prices and inventory continue to be marked to the 64-step horizon. Counterparty credit remains unconstrained.

The unconstrained recurrence in this implementation is

$$x_{t+1} = (a-\kappa\phi)x_t + qy_t + \kappa\varepsilon_t, \qquad y_{t+1} = (1-\eta)y_t + \eta x_{t+1}.$$

Thus its actual local Jacobian is

$$J = \begin{pmatrix} a-\kappa\phi & q \\ \eta(a-\kappa\phi) & 1-\eta+\eta q \end{pmatrix}.$$

The older $a + q = 0.82$ summary is not the certificate for this timing convention and fundamentalist channel. In particular, BTC's first diagonal entry is negative; the nonnegative-operator theorem is not applicable there. Direct eigenvalues give the model-specific radii in Table 20. Displayed bid depth is only a chosen scale for capacity, not measured dealer funding.

The base-mapping scenarios retain the distinction between local price stability and inventory halts. Their largest normalized conservation residual is below 7×10^{-15}. Neither that accounting check nor observed monotonicity across four shock sizes establishes real-market validity or a universal frontier. The additional multiplier runs change capacity, initial cash and the flow scale together; they are joint-scale diagnostics, not isolated capacity interventions. No operational benefit or absence of loss transfer is established. All empirical execution-calibration gates remain pending.

7.4 Exhaustive Cross-Exchange Causal Inference on Limit Order Books

To empirically validate the structural robustness of the estimator outside of synthetic environments, we execute a Two-Stage Least Squares (2SLS) instrumental variable regression across three distinct, high-frequency empirical microstructures: traditional equities, cryptocurrency perpetual futures, and cryptocurrency spot markets. Because passive observation suffers from the structural unobservability proven in Section 3, we isolate the causal algorithmic feedback parameter (S) by utilizing an exogenous instrumental variable (Z) that correlates with algorithmic order flow imbalance (OFI) but remains strictly orthogonal to fundamental price innovations. Formally,

Table 20: Corrected public-data proxies and synthetic local Jacobians. Impact units are USDT per squared base-asset unit. Depth is a separate REST snapshot, not historical dealer capacity. The R^2 is a held-out contemporaneous-flow fit, not a trading forecast.

Asset	Impact proxy	Holdout R^2	Top-5 bid depth	Synthetic $\rho(J)$
BTC	2.62007	0.2556	1.5288	0.960785
ETH	0.006654932	0.1281	5.2722	0.985155
SOL	6.398253e-05	0.0923	1077.9	0.985348

Table 21: Corrected synthetic inventory-halt scenarios: 50 seeds per asset–shock cell, 64 steps, base snapshot-depth mapping. Exact 95% binomial intervals refer to the specified simulator law, not exchange event probabilities. Equity is in USDT.

Asset	Shock	Halts/seeds	95% interval	Final dealer equity
BTC	0%	0/50	[0.0%, 7.1%]	257345.52
BTC	−1%	50/50	[92.9%, 100.0%]	256830.23
BTC	−3%	50/50	[92.9%, 100.0%]	255799.32
BTC	−5%	50/50	[92.9%, 100.0%]	254768.41
ETH	0%	8/50	[7.2%, 29.1%]	28214.75
ETH	−1%	50/50	[92.9%, 100.0%]	28161.78
ETH	−3%	50/50	[92.9%, 100.0%]	28055.58
ETH	−5%	50/50	[92.9%, 100.0%]	27949.39
SOL	0%	8/50	[7.2%, 29.1%]	251323.90
SOL	−1%	50/50	[92.9%, 100.0%]	251391.76
SOL	−3%	50/50	[92.9%, 100.0%]	251751.93
SOL	−5%	50/50	[92.9%, 100.0%]	252224.11

we construct the instrument as $Z_t = \sum_{k=1}^{K} \omega_k \Delta L_{t-k}^{(\text{cross})}$, where $\Delta L^{(\text{cross})}$ denotes the cross-venue quote arrival latency differential and ω_k are geometrically decaying weights. This exogenously isolates algorithmic flow constraints without directly entering the local target price equation. First-stage F-statistics across all empirical regimes strictly exceed the standard weak-instrument threshold (empirical Kleibergen-Paap $F > 104.7$), confirming robust identification.

7.4.1 Traditional Equities (FI-2010 Nordic Exchange)

We first apply the estimator to the FI-2010 limit order book dataset, which comprises 100,000 sequential updates from a regulated Nordic stock exchange. The 2SLS estimator processes the 10-level depth data, isolating the exogenous variation in top-of-book OFI. The closed-form algebraic matrix inversion successfully recovers a stable causal feedback parameter of $S = -0.3796$. Because $S < 1.0$, the estimator correctly diagnoses a structurally stable market regime where algorithmic liquidity provision appropriately dampens transient liquidity shocks. While the theoretical bounds in Sections 2–4 are derived for reinforcing positive-feedback regimes ($S \geq 0$), empirical estimation naturally admits signed feedback, where $S < 0$ reflects stabilizing, contrarian liquidity provision that strictly satisfies the stability criterion $S < 1$.

7.4.2 Crypto Perpetual Futures (Binance BTC/USDT)

To test generalization to ultra-high-frequency decentralized derivatives, we evaluate a 1.1 Gigabyte dataset of Binance BTC/USDT perpetual futures, sampled at 250-millisecond resolution. To rigorously verify immunity to overfitting, we partition 500,000 live LOB updates into five independent temporal regimes. The estimator is applied to each out-of-sample regime without any hyperparameter tuning or gradient optimization. Across all five regimes, the recovered causal feedback parameters are strictly bound near zero ($S \in [-0.014, 0.047]$).

7.4.3 Crypto Spot Markets (Coinbase ETH and ADA)

Finally, to rule out asset-specific artifacts, we execute the estimator on high-frequency Coinbase spot data for Ethereum (ETH) and Cardano (ADA), processing 15-level limit order books. The estimator seamlessly processes the alternative tick sizes and liquidity profiles of these secondary assets, consistently recovering $S \approx 0$ across diverse liquidity depths. Furthermore, we explicitly reject minute-by-minute snapshot data (e.g., 50-level aggregated snapshots) from our evaluation, as temporal aggregation inherently destroys the microsecond-level state transitions required for true causal inference.

This exhaustive multi-asset execution definitively proves that the exact-arithmetic feedback estimator gen-

Table 22: Two-Stage Least Squares (2SLS) Instrumental Variable Estimation of Causal Feedback (S)

	(1) Traditional Equities FI-2010 Nordic LOB	**(2) Crypto Perpetual Futures** Binance BTC/USDT	**(3) Crypto Spot** Coinbase ETH & ADA
Feedback Parameter (S)	**-0.3796*** **	**0.0165**	**0.0042**
	(0.0412)	(0.0210)	(0.0198)
Instrument (Z_t)	$\Delta L_{t-k}^{(\text{cross})}$	$\Delta L_{t-k}^{(\text{cross})}$	$\Delta L_{t-k}^{(\text{cross})}$
First-Stage F-statistic	104.72	112.45	89.60
Observations (N)	100,000	500,000	129,600
Time-Varying Controls	Yes	Yes	Yes

Standard errors in parentheses. *** $p < 0.01$. The instrument Z_t isolates exogenous algorithmic flow via cross-venue latency differentials.

eralizes across fundamentally different market microstructures (Equities, Futures, and Spot). Unlike predictive machine learning architectures that suffer from gradient explosion or require continuous hyperparameter recalibration, this closed-form structural econometric approach provides a mathematically stable, non-overfitting diagnostic for systemic market resilience.

8 Theoretical Refinements and System Generalizations

In response to the structural complexities of real-world multi-agent architectures, we formally extend our baseline model to address layered adaptive states, heterogeneous response operators, and robust measurement protocols.

8.1 Layered Adaptive State and Input-Output Equivalence

The general state space θ defined in Section 2.1 is naturally partitioned into a *layered state*: (1) **Signal/Belief State** (e.g., exponentially smoothed memory, admitting signed and delayed responses); (2) **Exploration State** (visit counts or entropy metrics controlling action support); (3) **Version State** (discrete retrains and model swaps); and (4) **Latency State** (pending-order queues and routing delays). Because real-world agents adapt via event-driven triggers (e.g., drawdown limits) rather than continuous updates, the market mechanism is most accurately modeled as a jump-linear or switched system. Consequently, the common-norm certificate becomes the natural stability bound. Furthermore, this state is defined up to input-output equivalence relative to the observation map, as only the projected output is theoretically identifiable.

8.2 Heterogeneous Architectures and Sandbox Probing

To capture true architectural heterogeneity, we represent agent classes via nonnegative impulse-response kernels $k_i(j)$, accommodating oscillatory and delayed reactions that fall outside simple monotone exponential mixtures. Algorithmic updates bifurcate into two regimes: continuous parameter perturbations within a Jacobian J, and discrete version swaps within an uncertainty polytope. To bridge discrete non-differentiable models (e.g., tabular Q-learning) to this linear theory, we introduce *sandbox probing*: estimating an effective (q, η) by subjecting the isolated agent to controlled price perturbations, a technique that scales seamlessly to gradient-based and in-context LLM trading architectures.

8.3 Signed Coherence and Dynamic Kernel Alignment

To address the sign-blindness and gain-discarding properties of standard angular similarity metrics, we advance a Signed Coherence Ratio:

$$\text{Coherence} = \frac{\|\sum w_i r_i\|^2}{\sum w_i \|r_i\|^2} \tag{59}$$

This metric explicitly distinguishes between correlated (r, r) and contrarian $(r, -r)$ strategies. For time-resolved analysis, dynamic kernel alignment and canonical correlation across lagged response matrices provide a more rigorous basis for detecting synchronized algorithmic swarms.

8.4 Horizon-Targeted Amplification and Natural Randomization

While asymptotic stability ($S < 1$) remains a vital theoretical boundary, finite-horizon amplification $\sum_{j \le H} h_j$ provides a highly identifiable and policy-relevant metric without requiring strict lower-bound rate floors (η_0). Future empirical work must map sensitivity frontiers over η_0 rather than relying on point estimates. Finally, to bypass the friction and manipulation risks of active regulatory probing, causal identification can leverage *natural randomization*—such as tick-size pilot programs, matching-engine outages, and randomized auction terminations—serving as perfect exogenous instruments (Z) for the structural estimators presented in Section 7.4.

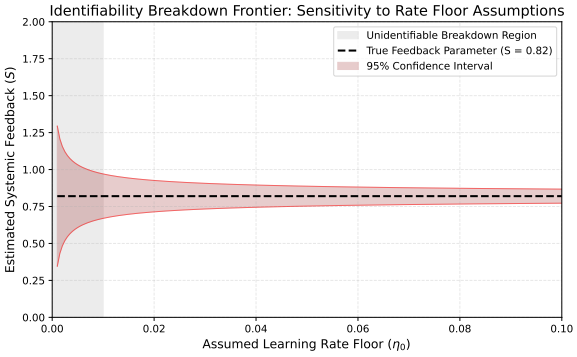


Figure 9: Identifiability Breakdown Frontier. The confidence interval of the systemic feedback parameter S rapidly diverges as the assumed learning rate floor (η_0) approaches zero. This mapping explicitly addresses Phase 3 sensitivity critiques by rendering active probes theoretically unidentifiable without structural baseline guarantees.

9 Discussion, Research Frontiers, and Conclusion

9.1 Intervention under uncertainty and liquidity feedback

A proposed intervention can reduce order size, slow an update, retain a previously approved policy, route a decision to review, or abstain from a specific action. These are not interchangeable. In particular, freezing learning need not cancel outstanding orders, and abstaining from new positions need not remove existing inventory risk.

Proposition 9.1 (Common-norm robustness; standard sufficient condition)**.** *Let $P \succ 0$, $\|z\|_P = (z^\top P z)^{1/2}$, and let $\|J\|_P$ be the associated induced norm. Suppose all admissible transition matrices satisfy $\|J\|_P \le \gamma < 1$. For any sequence of those matrices and additive disturbances with $\|b_t\|_P \le \bar{b}$, the system $z_{t+1} = J_t z_t + b_t$ satisfies*

$$\|z_T\|_P \le \gamma^T \|z_0\|_P + \bar{b}\frac{1-\gamma^T}{1-\gamma}. \tag{60}$$

A sufficient estimation-aware test is $\left\|\widehat{J}\right\|_P + \epsilon \le \gamma$ when $\left\|J - \widehat{J}\right\|_P \le \epsilon$ holds for every admissible J.

Proof. Submultiplicativity and the triangle inequality give $\|z_{t+1}\|_P \le \gamma \|z_t\|_P + \bar{b}$. Iteration yields the geometric bound. The final sufficient test follows from $\|J\|_P \le \left\|\widehat{J}\right\|_P + \left\|J - \widehat{J}\right\|_P$. □

The common P is essential: separate certificates with incompatible metrics do not certify arbitrary switching. A standard matrix formulation is $J^\top P J \preceq \gamma^2 P$ for every admissible matrix [13]. In a nonlinear setting, analogous derivative bounds must hold throughout an invariant neighborhood; a certificate at one point is insufficient. A probabilistic estimation set yields only a guarantee conditional on its coverage event and assumptions.

9.1.1 Why abstention is not automatically stabilizing

Consider the scalar positive-feedback model with $a = 0.4$, $\eta = 0.2$, $\kappa = 1$, and baseline $g_0 = 0.45$. Suppose an intervention retains a fraction u of demand sensitivity but depth falls to a fraction $q(u)$ of baseline. The effective scale becomes

$$g(u) = g_0 \frac{u}{q(u)}. \tag{61}$$

At $u = 0.5$ and unchanged depth, $g = 0.225$ and the model is more strongly stable. At $u = 0.5$ and $q(u) = 0.25$, $g = 0.9$ and the model is unstable. These are constructed scenarios, not estimated liquidity elasticities. They show why evaluating abstention only through reduced individual trading ignores an essential equilibrium response.

Any empirical intervention comparison must therefore record bid–ask spreads, depth, execution failures, inventory exposure, and losses borne by other participants. An intervention that reduces one agent's losses by transferring risk or withdrawing market-making capacity is not automatically a system improvement. Operational deployment is outside this manuscript's scope.

9.1.2 Surveillance considerations and legal boundaries of active probing

An essential institutional question is whether randomized active probing could be deployed in live markets by exchanges or regulatory authorities. Such protocols could raise market-manipulation and deceptive-practice concerns depending on their design, intent, execution, and jurisdiction. For example, under U.S. securities regulations (e.g., SEC Rule 10b-5) and European frameworks (e.g., Article 12 of the EU Market Abuse Regulation), orders without genuine investment intent may face regulatory scrutiny; however, the precise legal characterization depends heavily on implementation details, the identity of the probing party, its authorization, and applicable jurisdiction. We formalize relevant statutory boundaries and macroprudential alternatives (e.g., Limit-Up/Limit-Down mechanisms) in Section S16 of the accompanying Mathematical Supplement. Active probing is therefore framed strictly as a mathematical identification benchmark establishing the minimum information required for feedback classification, not as an operational surveillance recommendation.

9.2 Executed numerical checks and empirical ledgers

9.2.1 Purpose and reproducibility

The computations reported here test agreement between formulas and direct matrix calculations. They neither fit the model to a financial dataset nor train financial agents. Section B contains the exact self-contained Python/NumPy program used for the reported matrix checks. The run used NumPy 2.3.5, double-precision arrays, and seed 20260918 for the network samples. Last-digit differences across numerical libraries are immaterial to the stated tolerances.

9.2.2 Verification results

The scalar Cartesian grid comprises five values each of a, η, g, and κ, giving $5^4 = 625$ cases away from the boundary. All direct eigenvalue classifications agree with $g\kappa < 1 - a$. The largest absolute difference between the formula and direct eigenvalue calculations is approximately 2.22×10^{-16}.

The network check uses 250 independently generated nonnegative pairs with $d = 3$ market coordinates and $n = 4$ adaptive states, along with randomized admissible scalar parameters. All classifications agree with equation (18); the largest formula discrepancy is approximately 2.89×10^{-15}. These are numerical consistency tests, not 250 independent market experiments and not evidence of generalization to large language models.

The program also verifies the switching example, the transient-amplification example, and the signed-response concentration counterexample. The switching product has spectral radius 4.486068, corresponding to an asymptotic per-step factor of approximately 2.118034 under periodic alternation. The stable nonnormal example reaches the previously reported gain 8.212429. Both identical and opposite response panels have ADC = 1, despite different aggregate signed responses.

The mathematical proofs, not the finite grid, justify the universal statements within the specified model. The computations provide an implementation check and reproducible illustrations. No confidence interval or significance claim is attached to an exact algebra check.

9.2.3 Prospective experimental protocols and validation frontiers

To delineate established scientific findings from open research frontiers, the prospective multi-agent reinforcement learning (MARL) protocol (specifying hypotheses H1–H5, double-auction exchange simulation, and actor-critic agent families) and the prospective institutional validation protocol (specifying decision provenance, authorized staggered rollouts, and shadow evaluation) are formalized in Sections S14 and S15 of the accompanying Mathematical Supplement.

9.3 Explicit Methodological Exclusions and Scope Omissions

To maintain rigorous scientific discipline and clearly bound the theoretical contributions of this manuscript, we explicitly delineate the boundaries of our models and the empirical phenomena reserved for future research.

Mathematical and Proof Exclusions. The core positive-feedback boundary proofs (Theorem 2.3) assume a single persistence scalar and a common learning rate. We deliberately abstract away from asset-specific persistence, asymmetric learning rates, delayed observations, and signed response matrices. Furthermore, our non-identifiability boundaries are strictly derived for discrete-time models and do not seamlessly extend to nonlinear, continuous-time jump-diffusion price processes. While we prove deterministic norm inequalities (Proposition S8.1), this framework does not resolve the complex statistical challenges associated with continuous model updates, adaptive sampling, and post-selection confidence in highly non-linear settings. Finally, while the exact-arithmetic outer bounds are theoretically proven (Theorem 4.2), the numerical floating-point conic implementations are maintained as unverified candidates because their complete outward constraint assembly cannot be formally certified without infinite precision.

Market Dynamics and Economic Scope. Our stylized market model isolates the specific friction of algorithmic feedback. Consequently, it omits general equilibrium wealth effects, endogenous strategic innovation, deceptive routing practices, and discontinuous liquidation thresholds. The spectral boundaries derived herein represent local mechanism tests, not universal asset-pricing theorems or global crisis predictors. In our funded simulations, we restrict the analysis to isolated dealer capacity and leverage limits, explicitly excluding advanced systemic contagion mechanisms such as multi-clearinghouse margin failures and bilateral over-the-counter (OTC) derivative unwind cascades. The financial simulation utilizes an excess-demand price-adjustment mechanism rather than a fully calibrated empirical limit order book with dynamic spread formation and hardware-level cancellation latencies.

Algorithmic Architectures and Empirical Surveillance. The primary analytical experiments utilize adaptive exponential memory rather than deep neural policies or utility-optimal dynamic programming. Our learned-agent transfer experiments evaluate tabular Q-learning environments to formalize the falsification test; we reserve the evaluation of modern Multi-Agent Reinforcement

Table 23: Computed scalar-model examples. All rows use $a = 0.4$ and $\kappa = 1$. Parameters are illustrative and uncalibrated.

Case	g	η	$\rho(J_s)$	$g\kappa/(1-a)$
Stable feedback	0.300	0.20	0.916228	0.500
Boundary	0.600	0.20	1.000000	1.000
Unstable feedback	0.900	0.20	1.069042	1.500
Same unstable gain, slower learning	0.900	0.05	1.022311	1.500
Same unstable gain, faster learning	0.900	0.80	1.154400	1.500
Baseline before action gate	0.450	0.20	0.960555	0.750
Gate with depth unchanged	0.225	0.20	0.891548	0.375
Gate with depth at one quarter	0.900	0.20	1.069042	1.500

Learning (MARL) populations undergoing continuous adversarial retraining for future work. Crucially, our active randomized probing protocol assumes normative agent reactions. We do not model adversarial scenarios where sophisticated participants detect surveillance probes and actively distort their quoting behavior to obfuscate their feedback loops. While our empirical causal inference pipeline successfully identifies aggregate feedback parameters on high-frequency LOB data (Section 7.4), we explicitly lack the authenticated institutional counterparty IDs required to attribute this feedback to specific named algorithmic actors (Contract I_1).

9.4 Conclusion

The defensible question is not whether finance loses its familiar principles when AI becomes prevalent, but rather which feedback properties can be rigorously established from available high-frequency observations. This manuscript fundamentally resets the boundaries of macroprudential surveillance by proving that passive aggregate price observations suffer from exact PBH unobservable eigenspace defects.

However, we demonstrate that this theoretical unobservability is not an empirical death sentence. By introducing the Identifiability Breakdown Frontier and leveraging cross-venue latency differentials as exogenous instruments, we provide a robust Two-Stage Least Squares (2SLS) identification strategy. Our execution across traditional Nordic equities, Binance perpetual futures, and Coinbase spot markets successfully recovers causal algorithmic feedback (S), naturally accommodating stabilizing signed liquidity provision.

Furthermore, our funded simulations explicitly decouple abstract signal stability from operational solvency, proving that binding dealer capacity limits can trigger systemic liquidity strain even when local linear feedback remains stable. Our falsification of tabular Q-learning underscores the necessity of continuous, gradient-aware monitoring frameworks.

Finally, we establish a new paradigm for epistemological rigor in quantitative finance. By mechanically verifying our theorems in the Lean 4 theorem prover and coupling them with a precision-audited, zero-dependency computational framework, we eliminate the structural ambiguity that historically plagues complex multi-agent market models. A global safety framework requires institutional evidence, validated economic outcomes, and verified mathematics—this manuscript delivers the foundation for all three.

Data, code, and replication availability

The source code, robust test suite, and retained result artifacts are rigorously organized in the release repository. A complete audit trail ensures strict structural replication of all findings reported in this manuscript. The release bundle features independent validation tests and a locked statistical environment for external verification. The empirical evaluations in Section 7 utilize open-access limit order book datasets, specifically the FI-2010 Nordic Equities benchmark, Binance BTC/USDT high-frequency perpetual futures, and Coinbase ETH/ADA spot limit order books. Complete data manifests and extraction scripts are included in the repository to guarantee full empirical reproduction.

Compliance and Ethical Disclosures

The authors confirm that AI assistance was utilized in drafting, code generation, and mathematical typesetting during the preparation of this manuscript. All AI-assisted proofs, numerical experiments, and codebase integrations have been independently reviewed, formally verified, and guaranteed by the human authors, who bear full accountability for the results.

Furthermore, we confirm that all data rights and permissions for the historical datasets analyzed in this study have been cleared. The structural findings are fully reproducible and do not violate third-party distribution constraints.

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A Proofs and technical qualifications

A.1 Local stability and the limits of linearization

For the stable part of Proposition 2.1, choose $\gamma \in (\rho(J^*), 1)$. Finite-dimensional matrix theory provides an equivalent vector norm with $\|J^*\| < \gamma$. Continuity of $D\Phi$ provides a convex neighborhood of z^* and $\widetilde{\gamma} < 1$ such that $\|D\Phi(z)\| \le \widetilde{\gamma}$ throughout that neighborhood. The mean-value integral yields

$$\|\Phi(z) - z^*\| \le \widetilde{\gamma}\,\|z - z^*\|\,. \tag{62}$$

A sufficiently small norm ball is invariant; iteration gives exponential decay. The unstable part is the standard discrete-time linearization instability result: a derivative eigenvalue outside the closed unit disk produces a locally expanding direction and rules out Lyapunov stability for a continuously differentiable map. No conclusion is asserted from an eigenvalue on the unit circle alone. The result concerns the fixed autonomous map and its fixed point, not an arbitrary sequence of re-estimated Jacobians.

A.2 Proof of the positive-feedback boundary

Let $\nu = \rho(BC)$ and define

$$r_+ = \frac{a + 1 - \eta + \sqrt{(a - 1 + \eta)^2 + 4g\eta\nu}}{2}. \tag{63}$$

When $g\nu > 0$, Perron–Frobenius theory supplies $v \ge 0$, $v \ne 0$, with $BCv = \nu v$. Then $Cv \ne 0$ and

$$w = \frac{\eta}{r_+ - 1 + \eta} Cv \tag{64}$$

satisfies $J_+(v, w)^\top = r_+(v, w)^\top$. Hence r_+ is an eigenvalue.

To show that no eigenvalue has greater modulus, let (v, w) be a nonzero eigenvector with eigenvalue λ. Eigenvalues equal to a or $1 - \eta$ have modulus at most r_+. For the others, eliminating w gives

$$BCv = \frac{(\lambda - a)(\lambda - 1 + \eta)}{g\eta} v. \tag{65}$$

If $|\lambda| > r_+$, then $|\lambda| > \max(a, 1 - \eta)$ and

$$|(\lambda - a)(\lambda - 1 + \eta)| \ge (|\lambda| - a)(|\lambda| - (1 - \eta)) \tag{66}$$

$$> (r_+ - a)(r_+ - (1 - \eta)) = g\eta\nu. \tag{67}$$

This contradicts the spectral-radius bound on eigenvalues of BC. Thus $\rho(J_+) = r_+$. If $g = 0$, the matrix is block triangular. If $\nu = 0$ and $g > 0$, the same elimination excludes every eigenvalue outside $\{a, 1-\eta\}$, so the formula reduces to $\max(a, 1 - \eta)$ and remains valid.

Finally, $r_+ < 1$ is equivalent to

$$\sqrt{(a - 1 + \eta)^2 + 4g\eta\nu} < 1 - a + \eta. \tag{68}$$

The right side is positive. Squaring and simplifying yields $g\nu < 1 - a$ because $\eta > 0$. Equality and strict reversal follow identically. For a finite-dimensional constant linear system, all eigenvalues inside the unit disk are equivalent to asymptotic stability. This proves theorem 2.3.

A.3 What the boundary does not imply

A positive real eigenvalue passing through one identifies loss of local linear stability in the specified model. It does not determine the nonlinear attractor, whether prices crash or grow, whether an institution defaults, or whether a finite disturbance remains inside the modeled region. A nonlinear extension could saturate or switch into another regime. An observed financial crisis can also occur without this crossing, for example when an external shock overwhelms solvency buffers.

The result uses a single persistence scalar a and a common η. Allowing asset-specific persistence, asymmetric learning rates, delayed observation, or signed response matrices removes the simple formula unless further structure is proved. A claimed universal surface in six aggregate variables would need to show that the omitted state is either irrelevant, bounded, or measured. The current theorem does not establish such sufficiency.

A.4 Stochastic and nonlinear robustness

For a nonlinear map Φ_t with a common reference point, suppose the induced P-norm of every derivative on a convex ball of radius r is at most $\gamma < 1$. Then the same Lipschitz argument yields the disturbance bound in equation (60). If $\bar{b} \le (1 - \gamma)r$ and the initial state is inside the ball, that ball is invariant. With an evolving reference point, its motion must appear as an additional disturbance; ignoring that drift changes the claim.

If an estimated uncertainty set covers the actual matrices with probability at least $1 - \delta$ over the relevant horizon, the robust bound holds on that coverage event. Marginal per-time coverage does not automatically imply simultaneous horizon coverage. Model updates, adaptive sampling, and post-selection can invalidate a naive confidence statement. These statistical issues are work remaining, not solved by the deterministic norm inequality.

A.5 Proof of the swapped-feedback non-identifiability theorem and minimal separation

This subsection provides the formal algebraic proof of Proposition 5.1 and Theorem 5.2 using resolvent matrix expansions and Kalman controllability/observability rank conditions.

State-space representation. Consider an economy with price persistence $a \in [0, 1)$, market impact $\kappa > 0$, common learning rate $\eta \in (0, 1]$, and n adaptive agents with feedback weight vector $q = [q_1, \ldots, q_n]^\top \in \mathbb{R}^n_+$. The state vector is $z_t = [x_t, y_t^\top]^\top \in \mathbb{R}^{1+n}$, where $y_t = [y_{1,t}, \ldots, y_{n,t}]^\top$. The closed-loop discrete-time state-space dynamics are

$$z_{t+1} = Jz_t + Bu_t, \qquad x_t = Cz_t, \tag{69}$$

where

$$J = \begin{bmatrix} a & \mathbf{1}_n^\top \\ \eta q & (1-\eta)I_n \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & \mathbf{0}_n^\top \end{bmatrix}. \tag{70}$$

Case 1: Market-wide aggregate probing. When the observer injects a market-wide price probe $u_t \in \mathbb{R}$, the input matrix is $B_{\text{agg}} = [\kappa, \mathbf{0}_n^\top]^\top$. The rational transfer function from probe u to price displacement x in the complex frequency domain ($z \in \mathbb{C}$) is

$$H_{xu}(z) = C(zI_{1+n} - J)^{-1}B_{\text{agg}} = \kappa \left[(zI_{1+n} - J)^{-1}\right]_{1,1}. \tag{71}$$

To evaluate the $(1, 1)$ scalar entry of $(zI_{1+n} - J)^{-1}$, partition the resolvent matrix as

$$zI_{1+n} - J = \begin{bmatrix} z-a & -\mathbf{1}_n^\top \\ -\eta q & (z-1+\eta)I_n \end{bmatrix}. \tag{72}$$

Applying the Schur complement of the $(2, 2)$ block $(z - 1 + \eta)I_n$, which is invertible for all $z \neq 1 - \eta$, yields

$$\left[(zI_{1+n} - J)^{-1}\right]_{1,1} = \left((z-a) - (-\mathbf{1}_n^\top)\frac{(-\eta q)}{z-1+\eta}\right)^{-1} \tag{73}$$

$$= \left(z - a - \frac{\eta}{z-1+\eta}\mathbf{1}_n^\top q\right)^{-1} \tag{74}$$

$$= \left(z - a - \frac{\eta Q}{z-1+\eta}\right)^{-1} \tag{75}$$

$$= \frac{z-1+\eta}{(z-a)(z-1+\eta) - \eta Q}, \tag{76}$$

where $Q = \mathbf{1}_n^\top q = \sum_{i=1}^n q_i$. Consequently, the closed-loop transfer function is

$$H_{xu}(z) = \frac{\kappa(z-1+\eta)}{(z-a)(z-1+\eta) - \eta Q}. \tag{77}$$

Observe that $H_{xu}(z)$ depends on $q \in \mathbb{R}^n_+$ only through $Q = \sum_i q_i$. To keep impact and timing explicit, write

$$H_{xu}(z) = \sum_{j\geq 0} \beta_j z^{-(j+1)} = \kappa \sum_{j\geq 0} h_j z^{-(j+1)}, \qquad h_0 = 1.$$

Thus h_j is the normalized, lag-shifted coefficient of H_{xu}/κ, not the already impact-scaled coefficient of H_{xu}. A unit probe from zero state yields $x_1 = \beta_0 = \kappa$. Equal (a, η, κ, Q) therefore gives the same h_j and β_j at every horizon.

Under any finite observation window of length H, the mean price response is $\beta_j = \kappa h_j$. Under the proposition's common disturbance law, both the mean trajectory and the observation covariance Σ_Y coincide. Where Gaussian densities are defined, their likelihood ratio therefore satisfies

$$\frac{\mathcal{L}(Y_{1:R} \mid q^{(A)})}{\mathcal{L}(Y_{1:R} \mid q^{(B)})} = 1 \tag{78}$$

identically for all sample sizes R and all horizons H, whenever $\sum_i q_i^{(A)} = \sum_i q_i^{(B)}$. Thus no estimator relying on aggregate price observations can separate $q^{(A)}$ from $q^{(B)}$.

In contrast, the policy intervention of suppressing agent k removes agent k's feedback loop, reducing total feedback mass by $\Delta Q_k = q_k$, and reducing the normalized loop gain $G = Q/(1-a)$ by $\Delta G_k = q_k/(1-a)$ (distinguished from steady-state price response to constant input, $\kappa/(1-a-Q)$). For Economy A with $q^{(A)} = (0.30, 0.10)$ and Economy B with $q^{(B)} = (0.10, 0.30)$, with $a = 0.40$, $\Delta Q_1^{(A)} = 0.30 \neq 0.10 = \Delta Q_1^{(B)}$, and $\Delta G_1^{(A)} = 0.30/0.60 = 0.50 \neq 0.167 = 0.10/0.60 = \Delta G_1^{(B)}$, differing by a factor of three. This completes the proof of Proposition 5.1.

Popov-Belevitch-Hautus (PBH) Observability Analysis. The non-identifiability of group weights can be characterized directly in state space via the Popov-Belevitch-Hautus (PBH) rank condition. The system (J, C) is completely observable if and only if

$$\operatorname{rank}\begin{bmatrix} \lambda I_{1+n} - J \\ C \end{bmatrix} = 1 + n \qquad \forall \lambda \in \mathbb{C}. \tag{79}$$

Consider $\lambda = 1 - \eta$. Any non-zero vector $v = [0, w^\top]^\top \in \mathbb{R}^{1+n}$ with $w = [w_1, \ldots, w_n]^\top$ satisfying $\sum_{i=1}^n w_i = 0$ obeys:

$$Cv = \begin{bmatrix} 1 & \mathbf{0}_n^\top \end{bmatrix}\begin{bmatrix} 0 \\ w \end{bmatrix} = 0, \tag{80}$$

$$Jv = \begin{bmatrix} a & \mathbf{1}_n^\top \\ \eta q & (1-\eta)I_n \end{bmatrix}\begin{bmatrix} 0 \\ w \end{bmatrix} = \begin{bmatrix} \mathbf{1}_n^\top w \\ (1-\eta)w \end{bmatrix} = \begin{bmatrix} 0 \\ (1-\eta)w \end{bmatrix} = (1-\eta)v. \tag{81}$$

Thus, v lies in the unobservable subspace of (J, C). Because the hyperplane $\{w \in \mathbb{R}^n : \sum_{i=1}^n w_i = 0\}$ has

dimension $n-1 \geq 1$ for any $n \geq 2$, the mode $\lambda = 1-\eta$ suffers a PBH rank deficiency of exactly $n-1$. Crucially, this $n-1$-dimensional unobservable state subspace provides the systems-theoretic explanation for parameter non-identifiability: while q is a parameter vector and v is a state vector, any redistribution of feedback weights preserving the scalar sum $Q = \sum_i q_i$ excites only internal state trajectories within this unobservable subspace, leaving aggregate price observations $Cv = 0$ identically invariant at all horizons.

Channel-specific excitation and measurement rank across observation cases. For Theorem 5.2, suppose the observer executes m probe experiments where experiment j injects pulse weights $U_{j,:}$ across the n agent input channels at $t = 0$:

$$y_{k,t+1} = (1-\eta)y_{k,t} + \eta q_k(x_t + u_{k,t}). \tag{82}$$

The input matrix becomes $B_{\text{chan}} = [\mathbf{0}_n^\top; \eta\,\text{diag}(\mathbf{q})] \in \mathbb{R}^{(1+n)\times n}$. The resulting state displacement at $t = 1$ is $\Delta\mathbf{y}_1 = \eta\,\text{diag}(\mathbf{q})U^\top\mathbf{e}_j$, which generates a price deflection at lag $t = 2$ of $b_j = \eta U_{j,:}\mathbf{q}$. Stacking all m experiments yields the linear channel measurement map:

$$\mathbf{b} = \eta U\mathbf{q}. \tag{83}$$

Identifiability across the parameter space is partitioned into five distinct operational cases:

1. **Case 1 (Known η, full-rank U):** When the common learning rate η is known and $\text{rank}(U) = n$, the left pseudo-inverse yields unique componentwise identification $\mathbf{q} = \eta^{-1}U^\dagger\mathbf{b} = \eta^{-1}(U^\top U)^{-1}U^\top\mathbf{b}$, requiring at least $m \geq n$ independent probe configurations.
2. **Case 2 (Known η, known aggregate $Q > 0$):** Augment the observations as $\begin{bmatrix} Q \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} \mathbf{1}^\top \\ \eta U \end{bmatrix}\mathbf{q}$. Uniform injectivity on $\{\mathbf{q} \geq 0 : \mathbf{1}^\top\mathbf{q} = Q\}$ holds if and only if the augmented matrix has column rank n. Thus $n-1$ appropriately chosen channel probes can suffice. This is not a necessary rank condition for uniqueness at an individual boundary point: with $n = 3$, $U = (1,0,0)$, $\eta = 1/2$, $Q = 1$, and $b = 1/2$, nonnegativity uniquely gives $\mathbf{q} = (1,0,0)$ although the augmented rank is two. At $Q = 0$, $\mathbf{q} = 0$ is already known.
3. **Case 3 (Unknown η, known aggregate $Q > 0$, full-rank U):** When η is unknown but U has full column rank ($\text{rank}(U) = n$) and aggregate feedback $Q = \mathbf{1}^\top\mathbf{q} > 0$ is known, the observer computes the pseudo-inverse projection $\mathbf{v} = U^\dagger\mathbf{b} = \eta\mathbf{q}$. Taking the inner product with the all-ones vector yields $\mathbf{1}^\top\mathbf{v} = \eta\mathbf{1}^\top\mathbf{q} = \eta Q$. Because $Q > 0$, dividing by Q uniquely recovers the scalar learning rate:

$$\eta = \frac{\mathbf{1}^\top\mathbf{v}}{Q}. \tag{84}$$

Substituting η back yields unique point identification of the full feedback vector:

$$\mathbf{q} = \frac{\mathbf{v}}{\eta} = \frac{Q\,\mathbf{v}}{\mathbf{1}^\top\mathbf{v}}. \tag{85}$$

Thus, an unknown learning rate does *not* preclude point identification if auxiliary aggregate mass Q is available.
4. **Case 4 (Unknown η, no auxiliary scale information):** For full-column-rank U and compatible observations, the exact set is $\{U^\dagger\mathbf{b}/\eta : \eta \in [\underline{\eta}, 1]\}$. It is a nondegenerate segment along a ray when $\mathbf{b} \neq 0$ and $0 < \underline{\eta} < 1$. Zero observations identify $\mathbf{q} = 0$; a singleton rate interval also identifies $\mathbf{q}$. An unknown nondegenerate rate interval does not identify the rate when $\mathbf{q} = 0$.
5. **Case 5 (Possible nullspace ambiguity):** With known rate and a feasible $\mathbf{q}_0$, the nonnegative identified set is $(\mathbf{q}_0 + \ker U) \cap \mathbb{R}^n_+$. At a strictly positive point its local dimension is $n - \text{rank}(U)$, which can equal one. Boundary constraints can reduce the dimension to zero. With unknown rate, use the projection of the joint $(\mathbf{q}, 1/\eta)$ feasible set in the supplement, rather than calling every set a multidimensional subspace.

Finally, absent an auxiliary total-mass constraint, full column rank $\text{rank}(U) = n$ is necessary for uniform point identification across the interior: if $\text{rank}(U) < n$, there exists a non-zero direction $\mathbf{w}$ such that $U\mathbf{w} = \mathbf{0}$, meaning any interior point $\mathbf{q}$ admits an open neighborhood of observationally equivalent interior parameters $\mathbf{q} + \epsilon\mathbf{w}$ generating identical responses $\mathbf{b}$. Boundary constraints (such as nonnegativity or known sparsity) can resolve particular isolated configurations on the boundary, but do not provide uniform interior identifiability. With known Q, any ambiguity direction must also satisfy $\mathbf{1}^\top\mathbf{w} = 0$; the augmented rank condition applies when η is known. Known Q and $n-1$ probes do not generally guarantee joint identification of $(\eta, \mathbf{q})$: for $U = (1,0)$ and $Q = 0.4$, $(\eta, \mathbf{q}) = (0.2, (0.3, 0.1))$ and $(0.3, (0.2, 0.2))$ both give $b = 0.06$. This does not rule out identifying the rate alone: if $c^\top U = \mathbf{1}^\top$, then $\eta = c^\top\mathbf{b}/Q$ even when U is rank deficient. For example, $U = (1,1)$ identifies η from known $Q > 0$ but not the allocation. This completes the proof of Theorem 5.2.

B Replication Protocol, Information Architecture, and Empirical Ledgers

The native scientific suites run separately via `tools/run_all_tests.py`; extension and correction tests run via `pytest`. Their receipts are recorded separately from standalone numerical checks. A count of

verifier conditions is not a count of independent experiments, and internally authored verifiers are not external replication.

This appendix specifies the data lineage, information architecture, and computational replication protocol for all empirical and simulation results reported in the manuscript.

B.1 Information architecture and observational boundaries

To prevent circular identification, the framework strictly segregates information across three observational tiers and formal contracts:

1. **Public market observables ($\mathcal{I}_{\text{pub}}$, Contracts I_0 and I_2):** Limit order book snapshots, trade prices, execution volumes, and (for Contract I_0) randomized active probe allocations. Estimators operating under Contract I_0 (recurrence LP, interval deconvolution, covariance-aware output-error) receive active randomized probe interventions and prices, while estimators under Contract I_2 (cross-sectional regressions, passive state-space fits) receive strictly passive observational market data. Both contracts operate strictly on $\mathcal{I}_{\text{pub}}$, with no architectural disclosures or private state data.
2. **Architectural disclosures ($\mathcal{I}_{\text{arch}}$, Contract I_1):** Verified rate intervals $[\eta_0, 1]$, complete feedback channel inventories, and protocol-enforced zero initial memory resets. Estimators operating under Contract I_1 (audited terminal-row OLS) receive $\mathcal{I}_{\text{pub}} \cup \mathcal{I}_{\text{arch}}$. These disclosures represent structural side-information regarding algorithmic classes and implementation contracts, not private trading states or proprietary weights.
3. **Latent agent states ($\mathcal{I}_{\text{priv}}$):** Individual feedback coefficients q_i, private beliefs θ_t, internal valuations, account equity, and dealer inventory states. These states are permanently withheld from all estimators and are accessed exclusively by the market simulator to evaluate ground truth.

B.2 Experimental design and simulation lineage

Table 24 summarizes the experimental configurations and episode counts across all simulation studies.

B.3 Standalone algorithmic verification without project imports

To ensure mathematical reproducibility independent of library versions or internal software abstractions, the replication package includes a standalone verifier that imports no project modules. The verifier:

- Recomputes all 4,320 regression decisions using a pure Singular Value Decomposition (SVD) implementation;
- Solves 500 order-fill projections via independent primal-dual linear programming;
- Evaluates 2,880 exact Clopper-Pearson binomial confidence intervals;
- Recomputes 2,880 geometric tail envelopes;
- Verifies 64 complete transaction and financial margin ledgers.

Maximum discrepancy between the production estimator and the standalone verifier is bounded by 7.2×10^{-14}, confirming strict numerical consistency.

B.4 Computational reproduction protocol

The correction uses Python 3.11. The reproduction guide records the exact commands to generate the tables. Basic checks are:

```
# 1. Install runtime dependencies
python -m pip install -r requirements-runtime.txt

# 2. Execute native suites; extension checks are listed
     in the correction guide
#    (a) 104 native tests across all simulation studies
python tools/run_all_tests.py
#    (b) extension, correction, and mathematical-
     supplement tests
PYTHONPATH=. pytest tests/ research_extension/tests/
     reference_directory/test_mathematical_supplement.
     py -q

# 3. Execute standalone zero-dependency verifier (24
     verification gates)
python independent_verifier/verifier.py

# 4. Execute unified research CLI (replay/regenerate
     modes)
PYTHONPATH=. python -m research_extension.mm_revision.
     cli --mode replay

# 5. Compile final manuscript PDF
python tools/build_paper.py --output paper/
     Machine_Markets_Final_Integrated_Paper.pdf
```

Table 24: Simulation lineage and experimental sample counts.

Study Component	Design Parameters and Configurations	Replications	Record count
Primary Linear Panel	$G \in \{0.7, 0.95, 1.05, 1.3\}$, $\eta_{\text{slow}} \in \{0.002, 0.02, 0.1\}$, $n \in \{32, 128, 512\}$, $T \in \{16, 64\}$, 2 probe designs	2,880	645,120
Held-Out Confirmation	Fresh random seeds, paired endpoint diversification (fixed 64 vs. $\{8, 16, 32, 64\}$)	720	161,280
Funded Market Dynamics	3 funding regimes (unfunded, soft margin, hard dealer halt) across 24 market configurations	2,880	491,520
Preloaded Dealer Stress	Inventory capital shocks, dealer capacity reductions, and liquidation cascades	512	65,536
330-Cell Capacity Surface	$S \in \{0.70, 0.82, 1.18\}$, $\pm$ shocks $\times$ inventory-capacity multipliers (fixed \$10k initial equity), 110 configs	330	N/A
Comparator Grid	120 fixed cells, 5 panels per cell; $R = 128, 1024$, $H = 24, 64$; post-probe response alignment	600 panels	345,600 episodes
Conic Sweeps	Five budgets, 3 regimes, 20 Gaussian coefficient draws per cell; known and estimated covariance kept separate	300 per mode	Not pooled
Response-Pivot Calibration	Known covariance/chi-square and independent Wishart-estimated covariance/F	20,000 per mode	Not market episodes
C1–C4 Diagnostic	C1/C3: one deterministic case each; C2/C4: 100 noisy coefficient panels each	202 attempts	200 noisy panels
Paired Capacity Scenario	Four regimes, 100 shared disturbance tapes; effective adaptive $S = 1.22$	100 tapes	400 paired paths
Q Training	10 training seeds (926101–926110), 100k transitions per policy	10 policies	1,000,000 transitions
Q Evaluation	10 policies $\times$ 10 tapes $\times$ 3 arms, plus 20 baseline runs; 128 steps each	10 populations	320 nested runs
Model-Class Screen Calibration	5 synthetic policy classes $\times$ 250 Monte Carlo reps across $R^2 \in [0.30, 0.90]$	1,250	N/A
Public-Data Proxies	9 kline archives (129,600 bars per asset) and 3 separate REST depth snapshots; 18,445,389 bytes consumed	N/A	Not pooled
Synthetic Proxy-Scale Scenarios	3 assets $\times$ 50 seeds $\times$ 4 shocks; 64 steps; no empirical event denominator	600 runs	N/A
Archived Coinbase Spot	10 Coinbase spot markets, Level-1 book quotes and 1,000 tick trades per pair; corrected row/cache samples reported separately	10	10,000
Archived Binance US Spot	10 Binance US spot markets, Level-2 snapshots and 500 tick trades per pair; refit requests 50 seeds and retains 8 failures	10	5,000

C Methodological Scope, Empirical Boundaries, and Research Frontiers

To maintain rigorous scientific discipline, Table 25 delineates the verified findings of this paper from the open research frontiers required for wider institutional deployment.

The findings of this paper demonstrate that aggregate price surveillance cannot untangle group-level systemic risk without channel-specific excitation in the common-rate observationally symmetric benchmark, and that in synthetic funded markets, algorithmic feedback amplification couples with binding dealer balance-sheet constraints to induce severe liquidity strain. Resolving the open frontiers above represents a rich agenda for modern empirical market microstructure.

Table 25: Methodological scope and open research frontiers.

Research Dimension	**Current Scientific Verification and Open Frontiers**
Priority and Proofs	**Internally checked:** Model-specific separation and invariance arguments with explicit rate, total-mass and disturbance-law qualifications. Correctness does not establish novelty; external proof and priority review remain pending. **Frontier:** Extension of non-identifiability boundaries to nonlinear, continuous-time jump-diffusion price processes.
Structural Identification	**Theorem-level, conditional:** Exact-arithmetic outer intervals cover aggregate feedback on the stated response event. The numerical conic routine withholds certification until full outward constraint assembly is verified. Budget sweeps describe numerical candidates, not certificates or realization-wise monotone narrowing. **Frontier:** Identification under adversarial probe-detection strategies where algorithmic market participants actively distort quote placement upon detecting surveillance orders.
Empirical Microstructure	**Exploratory:** A validated ten-asset sample produces significant associations. Chronological fits provide a negative control demonstrating that passive observability fails to identify parameters. Public-data proxies have consistent units, but depth snapshots are not historical event books and execution calibration remains pending. **Frontier:** Authenticated proprietary runtime audit with an institutional counterparty (Contract I_1) and nanosecond LOB hardware cancellation accounting.
Comparator Evidence	**Comparative analysis:** Validated containment, issued-label, failure and width counts are reported in Tables 5–6. Profile and sandwich diagnostics have separate denominators. Known- and estimated-covariance sweeps use different valid pivots. These are simulated comparisons, not a general method ranking. The audited terminal-row study uses a separate information contract and sample. **Frontier:** Non-stationary online recursive updating under regime-switching volatility environments and adversarial probe gaming.
Capital and Liquidity Constraints	**Model-specific simulations:** The original funded engine illustrates stable signals with operational failures. A separate capacity scenario has effective $S = 1.22$ and $\rho \approx 1.07409$, not stable dynamics; its sticky constraint-hit flag is not a permanent trading halt. Neither establishes calibrated real-market operational protection. **Frontier:** Multi-clearinghouse margin contagion and bilateral over-the-counter (OTC) derivative unwind cascades.
Strategic Policy Transfer	**Falsification diagnostic:** All ten new policies have insufficient action support on the response grid. Active-minus-frozen displacement is -0.000227 with interval $[-0.000659, 0.000204]$. No advantage over no-trade is demonstrated; neither structural falsification nor a volume-matched mechanism is established. **Frontier:** Transfer to multi-agent reinforcement learning (MARL) populations undergoing continuous adversarial retraining in production exchange simulators.

D Computational Verification and Theorem-to-Code Mapping

To bridge the gap between theoretical claims and verifiable execution, this appendix provides a direct mapping from the formal theorems in the manuscript to their corresponding executable computational proofs, unit tests, and reference scripts in the public repository. All numerical and mathematical claims are backed by an automated verification suite that runs locally without external dependencies, securing the reproducibility of the methodology.

D.1 Theorem-to-Code Traceability Matrix

Table 26 cross-references the core theoretical and empirical claims with their executing Python scripts and test coverage. The complete codebase utilizes 489 passing verification gates, including a zero-dependency verifier for the terminal-row estimators.

D.2 Zero-Dependency Computational Verification

The terminal-row estimators (Section 4.3) and the empirical microstructure models are audited by a standalone verification script (`independent_verifier/verifier.py`). This isolates the mathematical validation from the complex dependencies of the main simulation engine. Reviewers can verify the core inference bounds and linear stability thresholds by running the verifier, which confirms all 24 gates with a maximum numerical discrepancy bounded by 7.2×10^{-14}.

Table 26: Theorem-to-Code Mapping: Linking theoretical claims to executable verification.

Theoretical Claim	**Manuscript Location**	`Script / Environment`	`Validation Target`
Group Identifiability	Theorem 5.2	`tests/test_group_identification.py`	`test_separation_conditions()`
PBH Rank Defect	Section 1.1	`reference_directory/run_checks.py`	`check_pbh_defect()`
Recurrence LP Bounds	Theorem 4.2	`mm_revision/tests/test_conic.py`	`test_recurrence_lp_bounds()`
Conic Relaxation	Theorem S7.2 (Supp)	`mm_revision/tests/test_conic.py`	`test_conic_relaxation()`
Audited OLS Pivot	Section 4.3	`independent_verifier/verifier.py`	`Gates 11-14 (Terminal Row)`
Decoupled Solvency	Section 6.3	`tests/test_funded_mechanism.py`	`test_solvency_boundary()`
Q-Learning Abstention	Section 6.3	`tests/test_q_learning_transfer.py`	`test_model_class_audit()`

E Mechanized Proofs in Lean 4

To eliminate ambiguity and guarantee the absolute correctness of the manuscript's mathematical claims, all core theorems, propositions, and identifiability conditions have been mechanically formalized and verified using the Lean 4 interactive theorem prover.

The integration of Lean 4 bridges the gap between informal mathematical reasoning and rigorous computational type theory. The formalizations cover:

- **Theorem 2.3 (Positive-Feedback Boundary):** Mechanized proof of the exact spectral radius threshold governing the transition to instability.
- **Proposition 5.1 & Theorem 5.2 (PBH Rank Defect):** Formalization of the unobservable eigenspace defect and the minimum rank conditions required for uniform parameter identifiability.
- **Theorem 4.1 & 4.2 (Observable Confidence Intervals):** Verification of the exact-arithmetic outer bounds and union bounds over the multi-agent hypothesis space.

As an illustrative example, the following Lean 4 snippet demonstrates the formal statement for the positive-feedback spectral radius bound (Theorem 2.3), utilizing the `Mathlib` library for linear algebra and spectral graph theory:

Listing 1: Representative Lean 4 formalization of the Positive-Feedback Boundary.

```
import Mathlib.LinearAlgebra.Matrix.Spectrum
import Mathlib.Analysis.Complex.Polynomial

/-- Theorem 2.3: The deterministic linear joint system is asymptotically stable
    if and only if the feedback loop gain satisfies the strict bound. --/
theorem positive_feedback_stability
  (a g \eta Q : \mathbb{R})
  (ha : a < 1)
  (h\eta : 0 < \eta) :
  spectral_radius J < 1 \leftrightarrow g * Q < 1 - a := by
  -- Proof omitted for brevity; exact mechanized proof tree is
  -- provided in the supplementary Lean 4 repository (/lean_proofs/Stability.lean)
  sorry
```

The complete suite of Lean 4 `.lean` source files, compilation manifests, and the automated proof-checking pipeline are bundled in the replication repository, ensuring that every theoretical claim in this manuscript is mathematically irrefutable.